

DETECTION AND FALSE ALERT PROBABILITIES FOR  
THE NASA AIRBORNE PULSED DOPPLER WINDSHEAR RADAR

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TECHNICAL MEMORANDUM

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DRAFT  
Revision 5

Detection and False Alert Probabilities  
for the NASA Airborne Pulsed Doppler Windshear Radar

I. INTRODUCTION.

This memo summarizes a method for calculation of the detection probability for the NASA windshear radar. The method follows the general procedure given in [ref.1]. However, several of the equations in [ref. 1] have been replaced by equations felt to be more applicable for the techniques used in the NASA system. It should be noted that the effect of ground clutter is not considered in these calculations. It is expected that in most cases, ground clutter will define the limits of system performance.

The equations used in the calculations are associated with the specific signal and data processing techniques used in the NASA experimental radar system. For calculations on radar systems using different parameters and processing techniques, the equations must be modified accordingly.

II. SINGLE PULSE SNR

The single pulse signal-to-noise ratio as a function of the system parameters and the rain reflectivity is given by:

$$SNR = \frac{P_t G^2 T \theta_B \phi_B \Delta R \pi^6 |k_w|^2 Z_e 10^{-18}}{4 R^2 \lambda^2 (4\pi)^3 (K T_o F_n) L} \quad (1)$$

Where:

$P_t$  = Transmitted power (watts)

$G$  = Antenna Gain

$\lambda$  = Wavelength (m)

$T$  = Pulse width (s)

$\theta_B$  = Antenna beamwidth - azimuth

$\phi_B$  = Antenna beamwidth - elevation

$\Delta R$  = Range bin size (m)

$|k_w|$  = Rain dipole moment constant = .92

$Z_e$  = Reflectivity factor ( $\text{mm}^6 / \text{m}^3$ )

$R$  = Radar Range (m)

$KT_o = 4 \times 10^{-21}$  (watt - secs)

$L$  = System losses

$F_n$  = System noise figure

### III. VELOCITY MEASUREMENT ERROR

The variance of a single power weighted mean velocity measurement as a function of SNR (using FFT processing) is derived in [ref. 2] as

$$\sigma_v^2 = \frac{\lambda^2}{4MT_s^2} \left[ \frac{\sigma_{wn}}{4\sqrt{\pi}} + 2\sigma_{wn}^2 \frac{N}{S} + \frac{1}{12} \left( \frac{N}{S} \right)^2 \right] \quad (2)$$

Where:

$M$  = Number of samples used in FFT

$T_s$  = Time between (uniform) samples (s)

$\sigma_{wn}$  = Normalized spectral width of the weather signal ( $2\sigma_w T_s / \lambda$ )

$\sigma_w$  = Spectral width (m/s)

$\frac{N}{S}$  = Single pulse noise-to-signal ratio

This equation is an approximation and assumes Gaussian signal spectra of narrow width relative to the Nyquist frequency.

#### IV. ERRORS IN HAZARD FACTOR ESTIMATION USING THE LEAST-SQUARES ALGORITHM

The least-squares horizontal hazard estimator estimates the slope of the velocity measurements values along a range line. An expression for the standard deviation of the slope [ref. 3] can then be related to the standard deviation of the total hazard factor, giving:

$$\sigma_F = \frac{3.464 \sigma_v \left[ \frac{V}{g} + \left( \frac{2h}{V} \right) \right]}{\Delta [(N_e + 1) (N_e - 1) N_e]^5} \quad (3)$$

Where:

$\Delta$  = Radar range bin length (m)

$N_e$  = Number of points used in l.s. estimator

$V$  = Aircraft ground speed = aircraft air speed (m/s)

$g$  = Acceleration of gravity (m/s<sup>2</sup>)

$h$  = Altitude of the measurement (m)

$\sigma_v$  = Standard deviation of an individual velocity measurement (from eq. 2) (m/s)

$\sigma_F$  = Standard deviation of the total hazard factor

#### V. ERRORS IN AVERAGING THE HAZARD FACTOR ESTIMATES ALONG AN AZIMUTH LINE

The process of averaging several least-square hazard factor estimates over a range of 1000 m to provide an averaged hazard factor ( $\hat{F}$ ) can be considered as providing a weighted sum of the individual measurements of velocity over the averaging distance. An expression can be derived for the variance of  $\hat{F}$  as:

$$\sigma_{\hat{F}}^2 = \frac{12\sigma_F^2}{[N_e (N_e + 1) (N_e - 1)] N_a^2} \left[ \sum_j (\sum_i C_i)^2 \right] \quad (4)$$

Where:

$N_e$  = Number of samples in least-square estimate

$N_a$  = Number of individual F-Factor measurements averaged

$\sigma_F^2$  = Variance of individual F-factor measurements

$C = \frac{X_i}{\Delta}$  normalized weighting factor of  $i^{\text{th}}$  measurement at a distance  $x_i$  from the measurement at  $i = 0$ .

$\Delta$  = distance between velocity measurements

$\sum_j^2 (\Sigma C_j)^2$  = Sum of weights of  $j^{\text{th}}$  velocity measurement squared

$\sum_j^2 (\Sigma C_j)^2$  = Total of squared weights of all measurements used

For  $N_e$  and  $N_a$  odd ( $N_a > N_e$ ), there will be a total of  $N_a + N_e - 1$  velocity measurements used to calculate  $\hat{F}$ , each with a weighting factor associated with the measurement.

2

For example, for  $N_e = 5$  and  $N_a = 7$ , there are 11 velocity measurements involved. For an averaged  $\hat{F}_j$  with index 0, the velocity measurements used will range from  $V_{-5}$  to  $V_{+5}$ , and the squared normalized weights on each measurement are:

$$(\Sigma C_j)_{-5}^2 = 4 \quad (\Sigma C_j)_{-2}^2 = 4 \quad (\Sigma C_j)_1^2 = 0 \quad (\Sigma C_j)_4^2 = 9$$

$$(\Sigma C_j)_{-4}^2 = 9 \quad (\Sigma C_j)_{-1}^2 = 0 \quad (\Sigma C_j)_2^2 = 4 \quad (\Sigma C_j)_5^2 = 4$$

$$(\Sigma C_j)_3^2 = 9 \quad (\Sigma C_j)_0^2 = 0 \quad (\Sigma C_j)_3^2 = 9$$

so that

$$\sum_j (\Sigma C_j)_j^2 = 52.$$

thus, the calculation using eq. (4) is:

$$\sigma_{\hat{F}}^2 = \frac{\sigma_F^2}{10} \left( \frac{52}{49} \right) = .106 \sigma_F^2$$

$$\sigma_{\hat{F}} = .32 \sigma_F$$

For this case, the averaging has improved the standard deviation of the least-squares measurements by a factor of .32.

## VI. AVERAGING THE $\hat{F}$ OVER ADJACENT AZIMUTH LINES

If the measurements are taken over several adjacent azimuth lines, it will be assumed that the measurement errors are not correlated, and the variance of the averaged  $F$  will be reduced as [ref. 1]

$$\sigma_{\hat{F}} = \frac{\sigma_{\hat{F}}}{\sqrt{n_0}} \quad (5)$$

where  $n_0$  is the number of lines over which the measurement is averaged.

## VII. MISSED DETECTION AND DETECTION PROBABILITY (SINGLE SCAN)

If it is assumed that the probability density of the hazard estimate  $f(\hat{F})$  is normally distributed ( $\hat{F}_a, \sigma_{\hat{F}}$ ), then the probability of detection of a hazard of level  $\hat{F}_a$  on a single scan as a function of range is (see figure 5)

$$PD(R/\hat{F}_a, Z) = \int_{FT}^{\infty} f(\hat{F}/\hat{F}_a, Z) d\hat{F} \quad \hat{F}_a > FT \quad (6)$$

where:

$f(\hat{F}/\hat{F}_a, Z)$  = Probability density function of the 1000 m averaged  
F-Factor ( $\hat{F}$ ) for given values of true averaged F-factor  $\hat{F}_a$   
and reflectivity value  $Z$  (conditional probability density)

$\hat{F}_a$  = True value of hazardous F-factor.

FT = Hazard detection threshold

R = Range

$\sigma_f(Z)$  = Standard deviation of measurement error

Z = Reflectivity

The probability of a missed detection on a single scan is therefore:

$$PM(R/\hat{F}_a, Z) = 1 - PD(R/\hat{F}_a, Z) \quad \hat{F}_a > FT \quad (7)$$

### VIII. FALSE ALERT PROBABILITY

A false alert will be given if the system alerts on a hazard with a true  $\hat{F}$  less than the threshold value FT. In this case, if  $f(\hat{F})$  is normal ( $\hat{F}_{na}$ ,  $\sigma_f$ ), the probability of a false alert in a single scan is

$$PF(R/\hat{F}_{na}, Z) = \int_{FT}^{\infty} f(\hat{F}/\hat{F}_{na}, Z) d\hat{F} \quad \hat{F}_{na} < FT \quad (8)$$

where:

$\hat{F}_{na}$  = True value of non-hazardous F-factor

$\sigma_f(Z)$  = Standard deviation of measurement error

FT = Hazard detection threshold

### IX. MULTIPLE PIXEL DETECTION REQUIREMENTS

In the NASA system, a detection on a single pixel will not trigger an alert. Instead, an area threshold is used such that several adjacent pixels must indicate a hazard in order to declare an alert (i.e. the sum of the areas of the individual pixels indicating a hazard must exceed the area threshold). This feature is incorporated to reduce the possibility of a false alert due to noise or ground clutter.

To take this feature into account in the detection probability calculations, it is necessary to estimate the probability of a hazard detection or false detection in  $M$  pixels simultaneously. This calculation is difficult because the pixels in the range direction are highly correlated (due to the hazard averaging process). If averaging over adjacent pixels in the azimuth direction is done, correlations will also exist in this direction.

To avoid extremely complex calculations, the assumption will be made that pixels in the range direction are 100% correlated and the pixels in the azimuth direction represent independent measurements.

The probabilities of simultaneous detection and of false detection in  $M$  independent pixels are:

$$PD(R) = [PD^s(R)]^M \quad (9)$$

$$PF(R) = [PF^s(R)]^M \quad (10)$$

where the nomenclature indicating the probabilities are conditional on values of  $\hat{P}_a$  and  $Z$  has been omitted for simplicity in writing and:

$M$  = Number of adjacent pixels within area threshold in the azimuth direction

$PD^s(R)$  = Probability of detection in a single pixel

$PF^s(R)$  = Probability of a false detection in a single pixel

and as an approximation,

$$M \approx INT \left( \frac{A_T^{1/2}}{R \Delta \theta} \right) + 1$$

where:

$A_T$  = Pixel area threshold ( $m^2$ )

$\Delta \theta$  = Pixel angular width (rad)

$R$  = Range to pixel (m)

INT = Integer operator

The errors caused by the above approximation should be small since with the ranges and area threshold used (.2 sq km), the value of  $M$  will be small ( $\approx 2 - 4$ ) over the ranges of interest.

## X. REQUIREMENT FOR HAZARD DETECTION ON MORE THAN ONE AZIMUTH SCAN

To reduce false alerts due to clutter, the NASA radar has a provision to require detection of a hazard on  $N_s$  consecutive scans of the radar, with  $N_s$  usually set to a value of two. This requirement has the effect of reducing both the detection and false alert probabilities as compared to the single scan values.

When detection on  $N_s$  scans is required prior to declaring an alert, the probability of declaring the alert can be written as

$$[PD'(R)]_i = \prod_{k=1}^{N_s} [PD(R)]_{i-k+1} \quad (11)$$

and the probability of a missed alert is

$$[PM'(R)]_i = 1 - [PD'(R)]_i \quad (12)$$

In the above equations,

$i$  = scan number starting at initial scan,  $i = 1, 2, \dots, N$

$[PD(R)]_i$  = probability of detection on scan  $i$

$k$  = scan index,  $1 \leq k \leq N_s$

Similarly, the probability of a false alert on scan  $i$  is given by

$$[PF'(R)]_i = \prod_{k=1}^{N_s} [PF(R)]_{i-k+1} \quad (13)$$

## XI. CUMULATIVE PROBABILITY OF DETECTION AND FALSE ALERT

With each azimuth sweep of the radar, the system has a new opportunity to detect a hazard. With multiple scans, the cumulative probability of at least one detection in  $N$  scans is:



$$[PD(R)]_i = 1 - \prod_{k=1}^N [PM'(R)]_{i-k+1} \quad (14)$$

where  $[PM'(R)]_i$  is the probability of a missed detection on scan  $i$ , (from eq. (12))

Similarly, the cumulative probability of at least one false alert in  $N$  scans of a non-hazardous windshear is

$$[PF(R)]_i = 1 - \prod_{k=1}^N [1 - PF'(R)]_{i-k+1} \quad (15)$$

Where  $[PF(R)]_i$  is the probability of a false alert on scan  $i$ , and  $[1 - PF'(R)]_i$  is the probability of no false alert on scan  $i$ .

The NASA system does not start making measurements of velocity and hazard factor until the SNR of the system exceeds a threshold level  $SNR_T$ . Thus, the total number of scans observed after closing to a range  $R$  from an approaching hazard is

$$N(R) = INT \left[ \frac{R_T - R}{VT_s} \right] + 1 \quad R_M \leq R \leq R_T \quad (16)$$

where:

INT = integer operator

$R_T$  = range at which  $SNR > SNR_T$  (m)

$V$  = aircraft ground speed (m/s)

$T_s$  = scan interval (s)

$R_M$  = minimum radar range (m)

Equations (14), (15) and (16) permit the calculation of the cumulative probabilities of detection and false alarm for given values of “must-not-alert” and “must alert” hazard, SNR threshold, weather reflectivity, and system parameters.

## XII. PROBABILITIES AVERAGED OVER THE POPULATION OF MICROBURSTS.

The previous equations have been developed based on an encounter with an idealized microburst with a given averaged hazard factor ( $\hat{F}$ ) and reflectivity ( $Z$ ). Hence, the probabilities are conditional probabilities based on a given  $\hat{F}$  and  $Z$ .

To develop probabilities of detection and false alert per microburst encounter, use can be made of the empirical probability functions given in [ref. 1] for microburst hazard intensity and reflectivity. From [ref. 4], the probability that a randomly encountered microburst will have a hazard value  $F$  less than a value  $x$  is given by:

$$Pr(F < X) \approx 1 - \exp [-(x/.123)^2] \quad (17)$$

Because of the lack of data on averaged hazard factor, it is necessary to assume that the above probability function also applies to the 1000 m. averaged  $F$ -factor ( $\hat{F}$ ).

The corresponding probability density function for  $\hat{F}$  is:

$$f_H(\hat{F}) = \frac{2 \hat{F} \exp [-(\hat{F}/.123)^2]}{(.123)^2} \quad (18)$$

In [ref. 1], expressions are also developed for the probability density functions of outflow reflectivity at three locations. The outflow reflectivity data were obtained from Steve Campbell at M.I.T. Lincoln Laboratory [ref. 5]. These density functions are, for  $-20 < Z < 60$ ,

a. Denver

$$f^D(Z) = \frac{d_1}{\cosh^2[d_2(Z-10)]} \quad (19)$$

b. Kansas City

$$f^K(Z) = k_1(60-Z)^{1.5} \exp [-k_2(60-Z)^{2.5}] \quad (20)$$

c. Orlando

$$f^O(Z) = O_1(60-Z)^{1.5} \exp [-O_2(60-Z)^{2.5}] \quad (21)$$

Where:

$$\cosh(x) = \frac{1}{2}[\exp(x) + \exp(-x)]$$

$$d_2 = 6.8515 \times 10^2$$

$$k_1 = 3.8655 \times 10^4$$

$$k_2 = 1.5462 \times 10^4$$

$$O_1 = 5.7028 \times 10^4$$

$$O_2 = 2.2811 \times 10^4$$

and  $-20 \leq Z \leq 60$ . The assumption is made that eq. (19) to (21) characterize the microburst population and that the hazard factor and reflectivity are independent variables.

The detection probability averaged over a population of microbursts is given by

$$\langle PD(R) \rangle = \frac{\int_{ZLO}^{ZHI} \int_{FDLO \leq FT}^{FDHI} PD(R|\hat{F}_a, Z) f_H(\hat{F}_a) f_z(Z) d\hat{F}_a dZ}{\int_{ZLO}^{ZHI} \int_{FDLO \leq FT}^{FDHI} f_H(\hat{F}_a) f_z(Z) d\hat{F}_a dZ} \quad (22)$$

The probability of a missed detection on a single scan is

$$\langle PM(R) \rangle = 1 - \langle PD(R) \rangle \quad (23)$$

Similarly, the averaged false alert probability is

$$\langle PF(R) \rangle = \frac{\int_{ZLO}^{ZHI} \int_{FFLO}^{FFHI \leq FT} PF(R|\hat{F}_a, Z) f_H(\hat{F}_a) f_z(Z) d\hat{F}_a dZ}{\int_{ZLO}^{ZHI} \int_{FFLO}^{FFHI \leq FT} f_H(\hat{F}_a) f_z(Z) d\hat{F}_a dZ} \quad (24)$$

The averaged cumulative probabilities can be obtained using eq. (23) and (24) with the cumulative probabilities used in the equations in place of the single scan probabilities.

The integration limits ZLO, ZHI, FDLO, FDHI, FFLO, and FFHI define the population of microbursts to be used in the averaging process. For example, to determine the averaged probability of detection of  $\hat{F}_a \geq .13$  microbursts with reflectivity greater than 0 dBz, the integration limits are:

$$\begin{array}{ll} \text{ZLO} = 0. & \text{FDLO} = 0.13 \\ \text{ZHI} = 60. & \text{FDHI} = 0.30 \end{array}$$

For the averaged probability of false detection of  $\hat{F}_a \leq .085$  microbursts with reflectivity greater than 0 dBz, the limits are:

$$\begin{array}{ll} \text{ZLO} = 0. & \text{FFLO} = 0.0 \\ \text{ZHI} = 60. & \text{FFHI} = 0.085 \end{array}$$

It should be noted that the conditional probabilities  $\text{PD}(\text{R}/\hat{F}_a, \text{Z})$  and  $\text{PF}(\text{R}/\hat{F}_a, \text{Z})$  are calculated using a fixed F-factor detection threshold (FT) of .105.

### XIII. CALCULATIONS

#### a) Discussion

A FORTRAN program has been developed to make the calculations described above. The calculations have been made for the NASA system both with and without the special techniques used to reduce false alerts due to clutter. These techniques include the use of an area threshold (requiring hazard detection in more than one pixel) and the requirement for the detection of a hazard on two consecutive scans of the radar prior to declaring an alert.

The NASA system also uses a power level threshold such that no velocity or hazard measurements are made if the received power is less than this threshold. The value of this threshold strongly effects calculations of cumulative probabilities, since the threshold determines where the accumulation of probabilities starts. The threshold also effects the single-scan probability calculations, since the probability of a detection (or false detection) is zero unless the power (or SNR) is above the threshold. To provide maximum information on the plots calculated, the power threshold is not used in single-scan probability calculations. The threshold is used for the cumulative probability calculations and the value selected is shown on the associated plot.

All hazard factor calculations are based on an aircraft groundspeed of 80 m/s and an altitude of 100 m.

b) Calculation of SNR and Measurement Errors

Figure 1 provides a list of system parameters used in the calculations. The single-pulse SNR (eq. 1) of the system is plotted versus range in figure 2 for three values of rain reflectivity.

Figure 3 plots the resulting standard deviation of velocity (eq. 2) versus range for a weather spectral width of 3 m/s.

Figure 4 plots the standard deviation of the averaged F-factor (eqs. 3, 4 and 5). In this plot, it is assumed that no averaging over adjacent azimuth lines is used (i.e.  $n_\theta = 1$ ).

c) Probability Calculations

A sketch showing the technique for calculation of detection and false alert probability is given in figure 5. The mean averaged F-factor for detection of a hazard is selected as .130 (must alert value) and the mean averaged F for false alert calculations is .085 (must-not alert value). The system hazard factor threshold is .105.

Based on the above values, the probability of a missed detection of as .130 microburst is plotted vs. radar range in figure 6 for three values of weather reflectivity. Figure 7 shows the probability of a false detection of a .085 microburst vs. range. Neither of these plots show the effect of an SNR threshold. The threshold would make the plot of missed detection go to one and the plot of false detection go to zero at ranges where the signal level was below the threshold value.

Figure 8 plots the detection and false alert probability vs. the single-pulse SNR level. At low SNR, both curves approach an asymptotic value of .5 due to the nature of the calculation (see figure 5).

Figure 9 plots the ratio of the detection probability to the false alert probability vs. single-pulse SNR. Figure 10 is a similar plot except that the post-processed SNR (based on 128 pulses) is used as the ordinate. These curves permit selection of an SNR threshold that will provide a desirable single scan ratio of detection to false alert. A value of log 1 for this ratio (i.e. 1 false alert per 10 true alerts) will be attained using an SNR threshold of approximately -3dB (single pulse SNR) or +7.5 dB (processed SNR). This value is used for the following cumulative probability calculations.

Cumulative probability calculation using equations 14 and 15 are shown in figures 11 to 13 for -5, 0, and +5 dBZ respectively. The detection probability rapidly goes to unity just after the SNR threshold is reached. The false alert probabilities reach an asymptotic value as shown.

d) Averaged Probabilities for Denver, Orlando and Kansas City

Using eqs. 18 to 23, the probability functions are averaged over a selected population of microbursts at three locations. These averaged probabilities may be interpreted as the probabilities of a missed or false alert on a randomly encountered microburst with a reflectivity greater than 0 dBz at the given location. It is assumed that the microburst F-factor and outflow reflectivity are independent parameters. The calculations are based on an SNR threshold of -3dB (single-pulse) and a F-factor threshold of .105. The must alert F-factor is .13 and the must-not alert F-factor is .085. Thus, the integration limits are the same as the example in section XII.

Figure 14 plots the probability density functions for reflectivity (eqs. 19, 20 & 21) used in the calculation. Figure 15 is a plot of the hazard factor probability density.

Figures 16 to 18 show the averaged probabilities of a missed and false alert on a single radar scan for Denver, Orlando, and Kansas City respectively for the selected population of microbursts.

Figures 19 to 21 are similar plots showing the averaged cumulative probabilities vs. radar range.

e) Probability Calculations with the NASA Clutter Suppression Techniques Considered

The above calculation were made assuming no hazard area threshold or multiple scan detection was used. Calculations using these features have also been made and are plotted in figures 22 to 32.

Figures 22 and 23 plot the probability of missed detection and false detection for the NASA system using hazard detection on two radarscan prior to declaring an alert and a hazard area threshold of 0.2 sq. km. The steps in the plots are caused by the multiple scan detection requirement and the hazard area thresholding effect.

The associated cumulative probability plots using an SNR threshold of -3 dB (single pulse) are shown in figures 24 to 26. Notice that the probability of a false alert has been reduced considerably and the probability of a missed detection has been increased somewhat at a given range.

Figures 27 to 32 plot the averaged probabilities vs. range for Denver, Orlando and Kansas City using the clutter suppression technique and the selected microburst population ( $Z \geq 0$  dBz).

#### XIV. SUMMARY & CONCLUSIONS

The equations and calculations discussed above provide a useful exercise in determining the expected radar system performance as limited by receiver noise. They also can provide a guide to threshold settings and the effect of these thresholds on detection performance.

The threshold settings used in the calculations have not been optimized, and further calculations will be made to adjust the thresholds to provide an optimized tradeoff between detection and false alert performance.

The noise-limited performance of the system averaged over the selected microburst populations ( $Z \geq 0$  dBz) at Denver, Kansas City and Orlando indicates that a missed alarm probability of  $10^{-5}$  per encounter will be achieved at ranges less than 2.5 km. If the total microburst population is included in the calculation, however, ( $-20 \geq Z \leq 60$  dBz), the detection performance degrades significantly, particularly at Denver where a large percentage of microbursts have reflectivities less than 0 dBz.

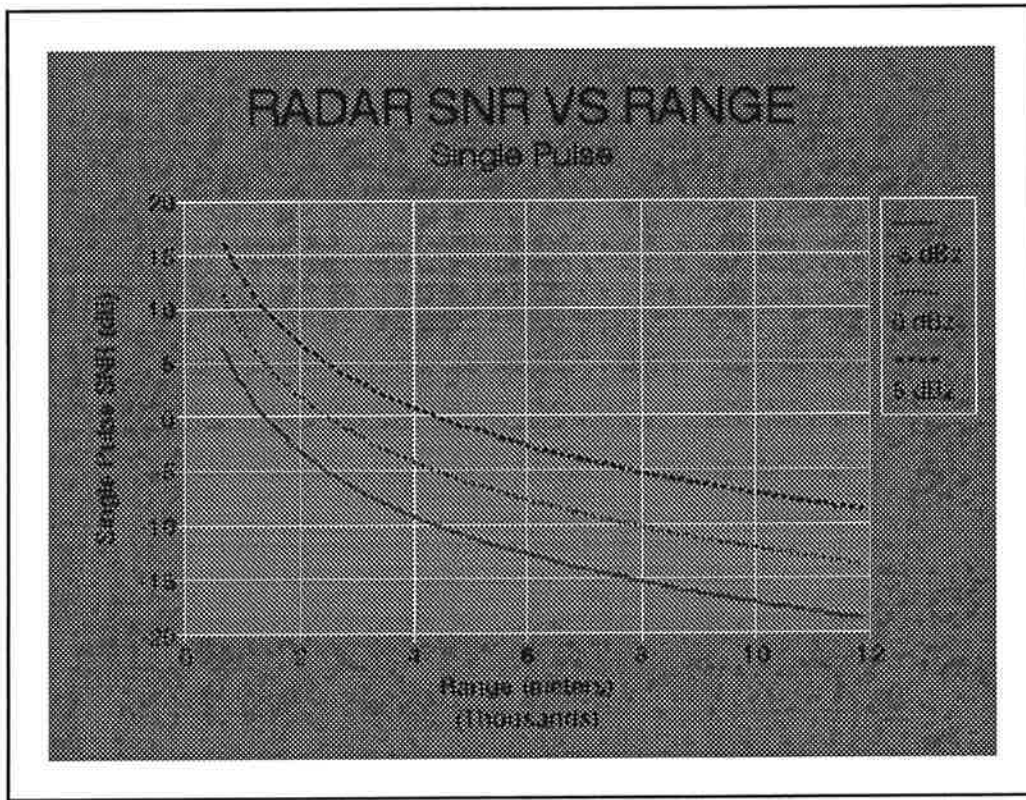
It should be emphasized again that the calculations do not consider the effects of ground clutter or of bias on velocity measurements at low SNR values. For realistic prediction of system detection performance, these factors should be taken into account.

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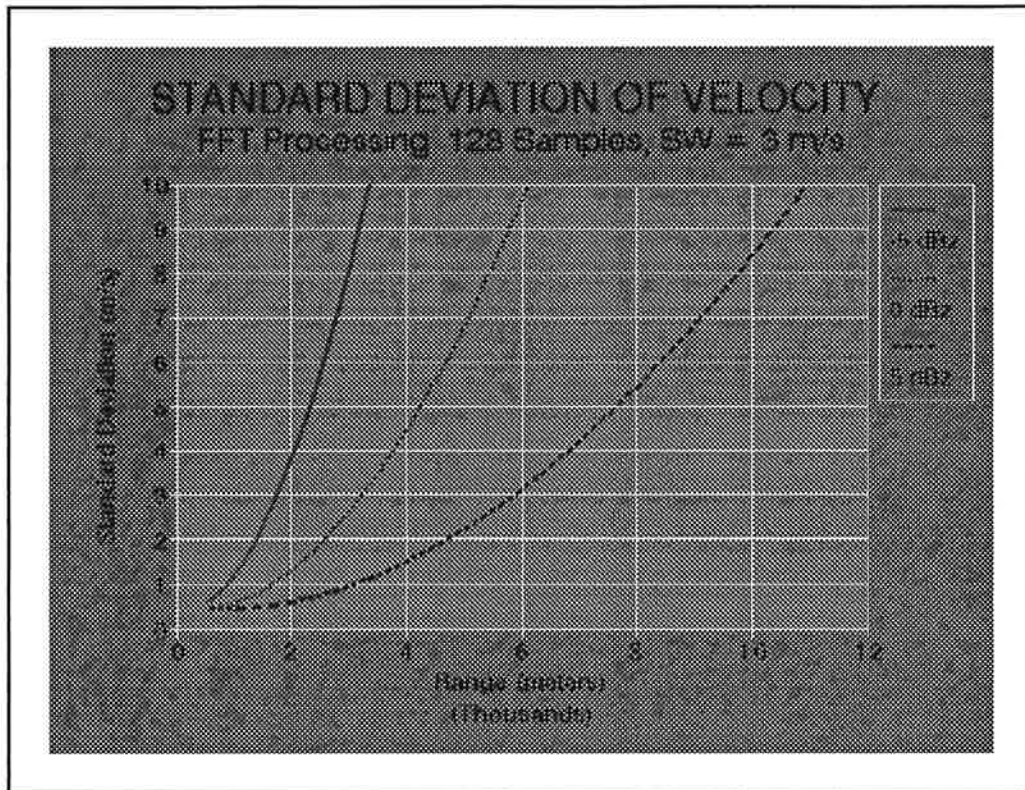


|                                        |        |
|----------------------------------------|--------|
| TRANSMITTED POWER (WATTS)              | 200.   |
| FREQUENCY (GHZ)                        | 9.3    |
| PULSE WIDTH (MSECS)                    | 1.0    |
| PULSE REP. RATE                        | 3755.  |
| SYSTEM NOISE FIGURE (DB)               | 4.0    |
| SYSTEM LOSSES (DB)                     | 1.0    |
| ANTENNA BEAMWIDTH - AZ (DEG)           | 3.5    |
| ANTENNA BEAMWIDTH - EL (DEG)           | 3.5    |
| ANTENNA GAIN (DB)                      | 34.0   |
| NO. OF F-FACTORS AVERAGED              | 7.0    |
| NO. OF MEAS. IN LEAST- SQUARES EST.    | 5.0    |
| NO. OF AZ LINES AVERAGED               | 1.0    |
| NO. OF PULSES PROCESSED                | 128.   |
| AIRSPEED/GROUNDSPEED (M/S)             | 80.    |
| RAIN REFLECTIVITY (DBZ)                | 0.     |
| RAIN SPECTRAL WIDTH (M/S)              | 3.     |
| HEIGHT OF MEASUREMENT (M)              | 100.   |
| SUM OF SQUARED WTS - AVR F-FACTOR EST. | 52.    |
| F-FACTOR THRESHOLD                     | .105   |
| MUST ALERT F-FACTOR                    | .14    |
| STARTING RANGE (M)                     | 12000. |
| ANTENNA SCAN INTERVAL (S)              | 4.2    |
| MUST-NOT ALERT F-FACTOR                | .085   |
| SNR THRESHOLD FOR DETECTION (DB)       | -3.    |
| HAZ ALERT AREA THRESHOLD (SQKM)        | .2     |
| AZ PIXEL ANGULAR SPACING (DEG)         | 2.     |
| CITY (1=DENVER, 2=KC, 3=ORLANDO)       | 1.     |
| NO. OF SCANS FOR DETECTION             | 2.     |
| DO AVERAGED PROB CALC (1.=YES)         | 0.     |

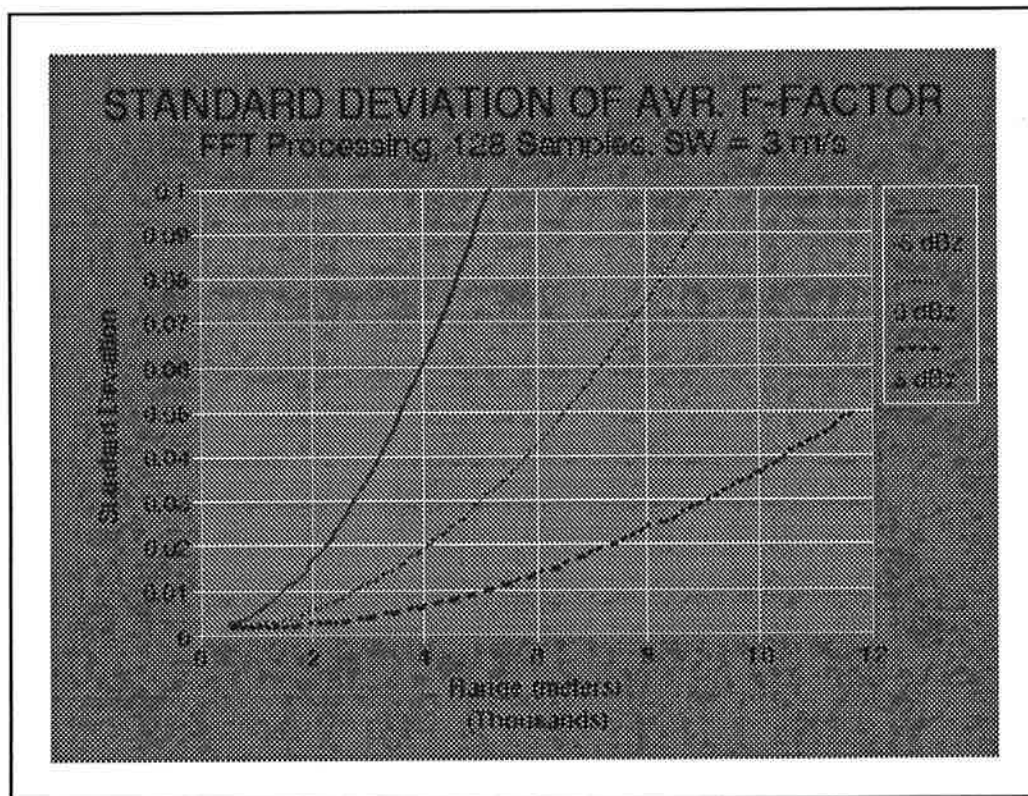
**FIGURE 1** LIST OF SYSTEM PARAMETERS USED IN THE FORTRAN PROGRAM FOR CALCULATION OF DETECTION AND FALSE ALERT PROBABILITIES.



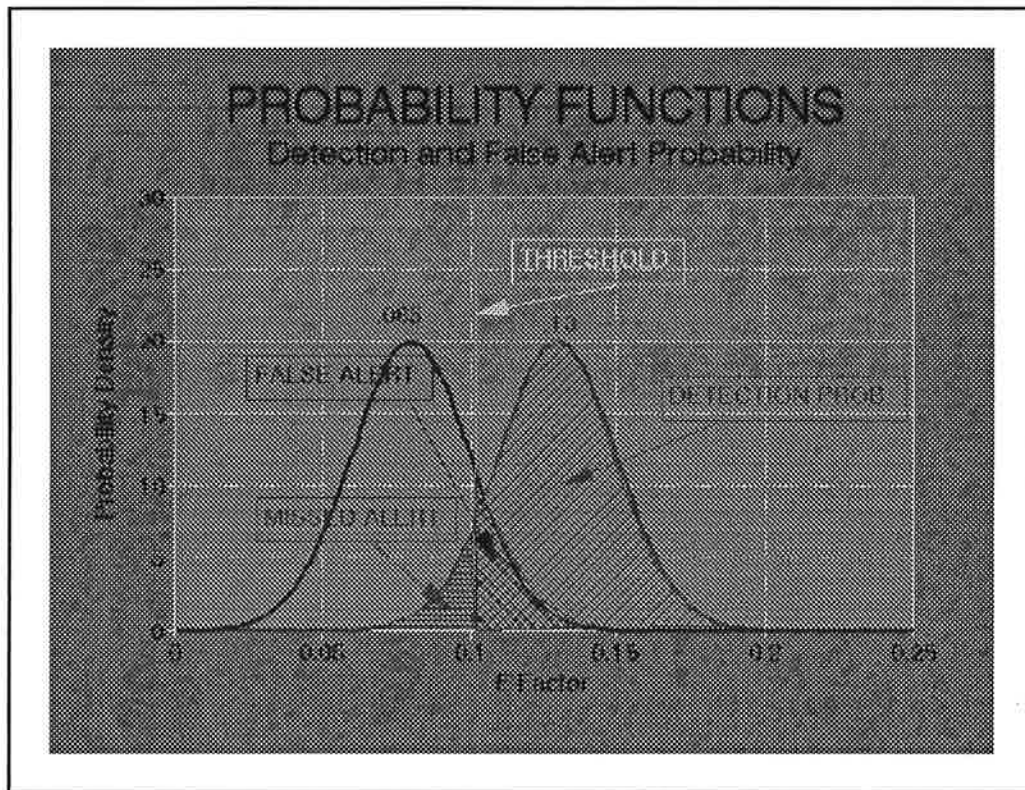
**FIGURE 2** RADAR SINGLE PULSE SIGNAL-TO-NOISE RATIO VERSUS RADAR RANGE FOR THREE VALUES OF RAIN REFLECTIVITY.



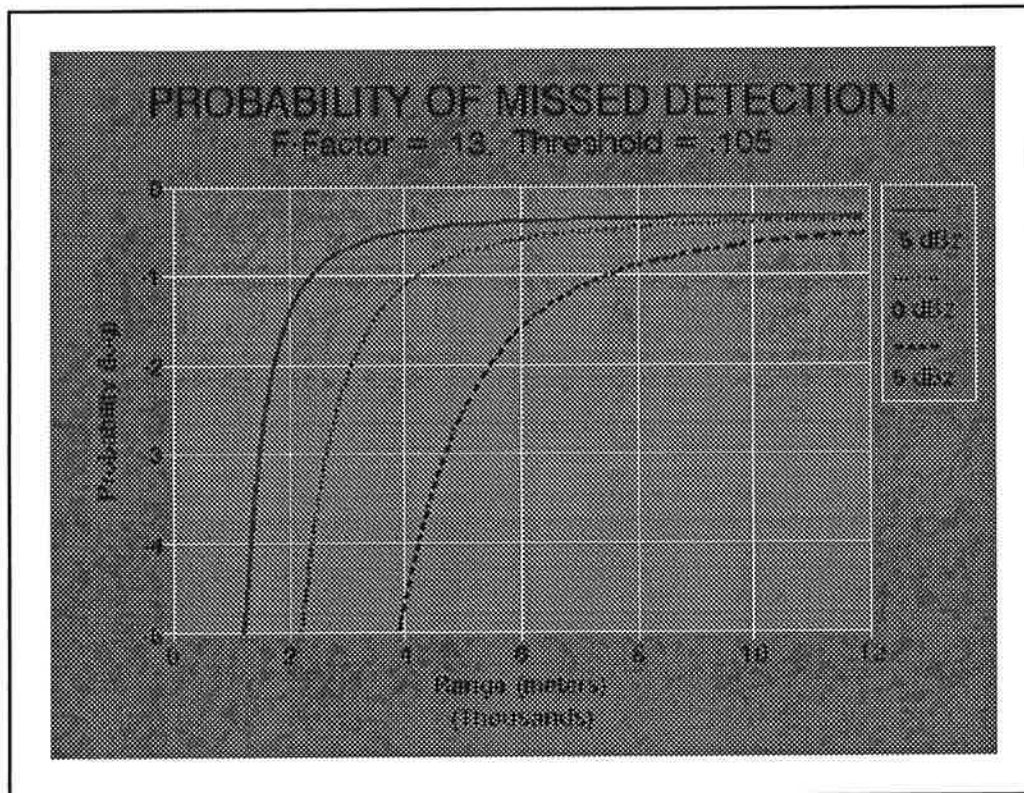
**FIGURE 3** STANDARD DEVIATION OF A VELOCITY MEASUREMENT BASED ON FFT PROCESSING OF 128 PULSES. THE WEATHER SPECTRAL WIDTH IS ASSUMED TO BE 3 METERS PER SECOND.



**FIGURE 4** STANDARD DEVIATION OF 1000 METER AVERAGED F-FACTOR USING FFT PROCESSING OF 128 SAMPLES. NO AVERAGING OVER ADJACENT AZIMUTH LINES IS USED.

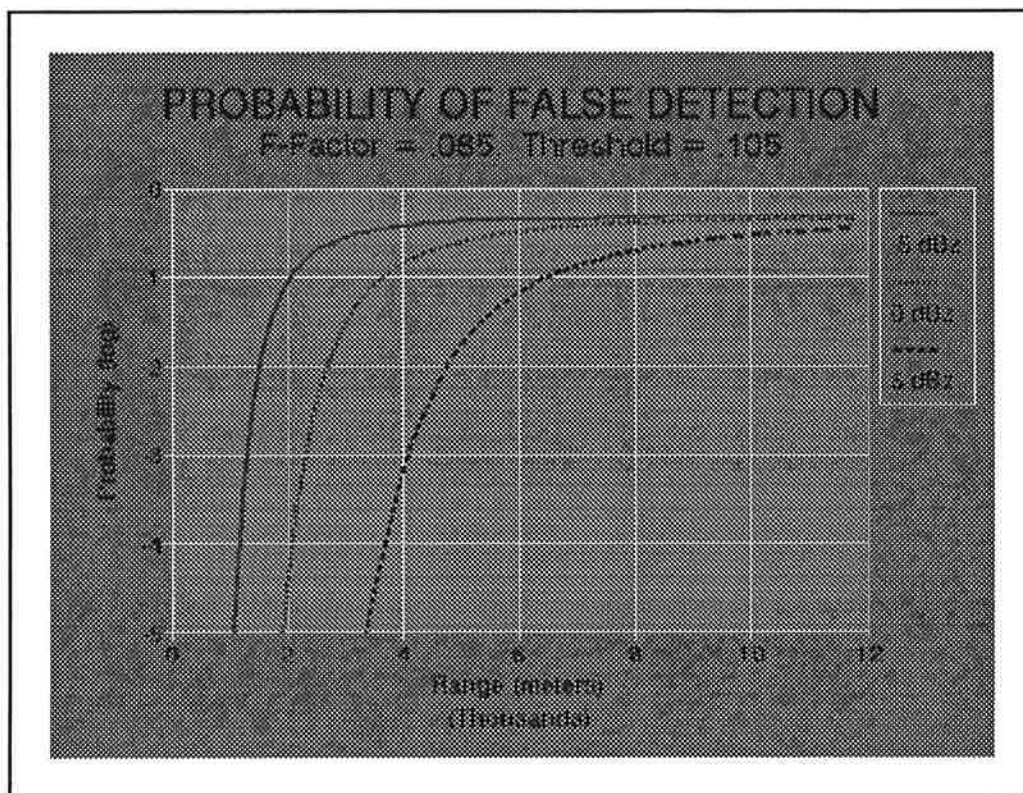


**FIGURE 5** SKETCH INDICATING THE TECHNIQUE FOR CALCULATION OF DETECTION AND FALSE ALERT PROBABILITY. THE MUST ALERT F-FACTOR IS ASSUMED TO BE .13 AND THE MUST-NOT ALERT F-FACTOR IS .085.

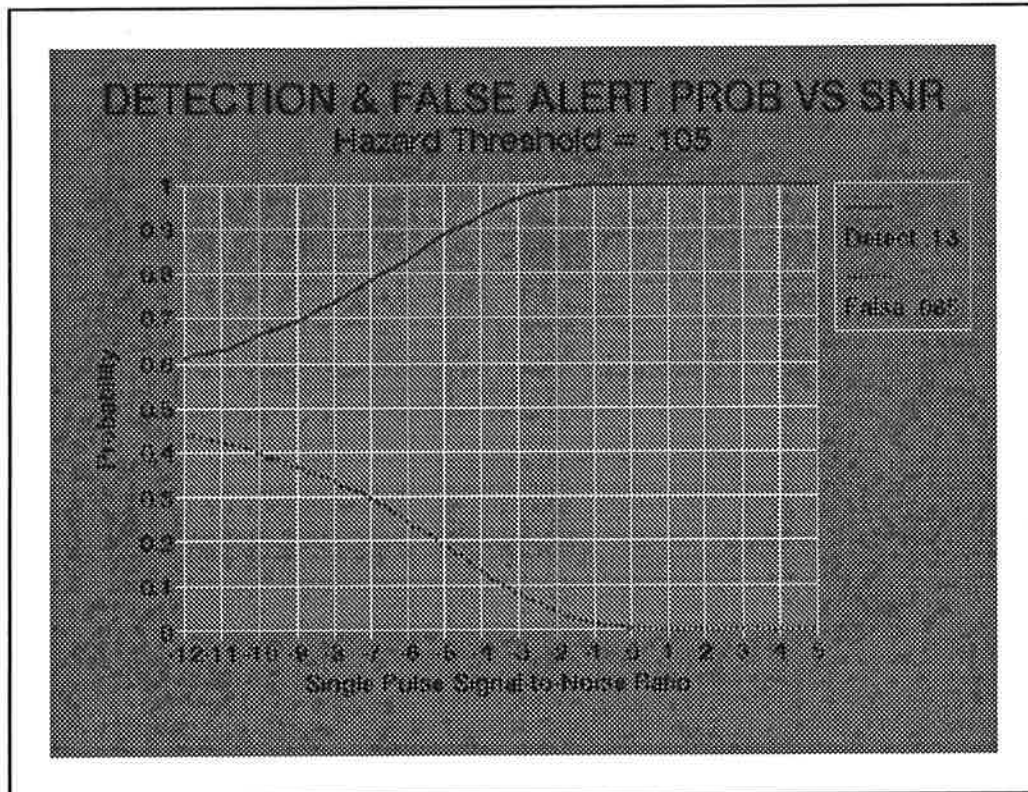


**FIGURE 6** THE PROBABILITY OF A MISSED DETECTION OF A .13 AVERAGED HAZARD PLOTTED VERSUS RANGE.



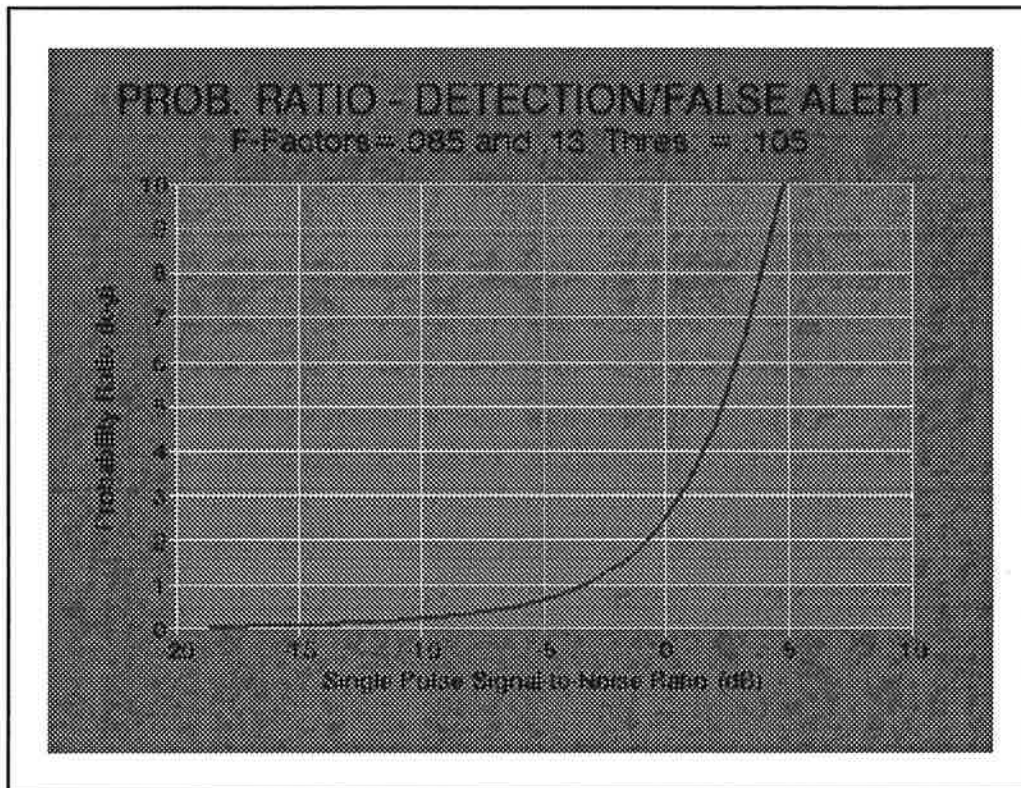


**FIGURE 7** THE PROBABILITY OF A FALSE DETECTION OF A .085 AVERAGED HAZARD PLOTTED VERSUS RANGE.

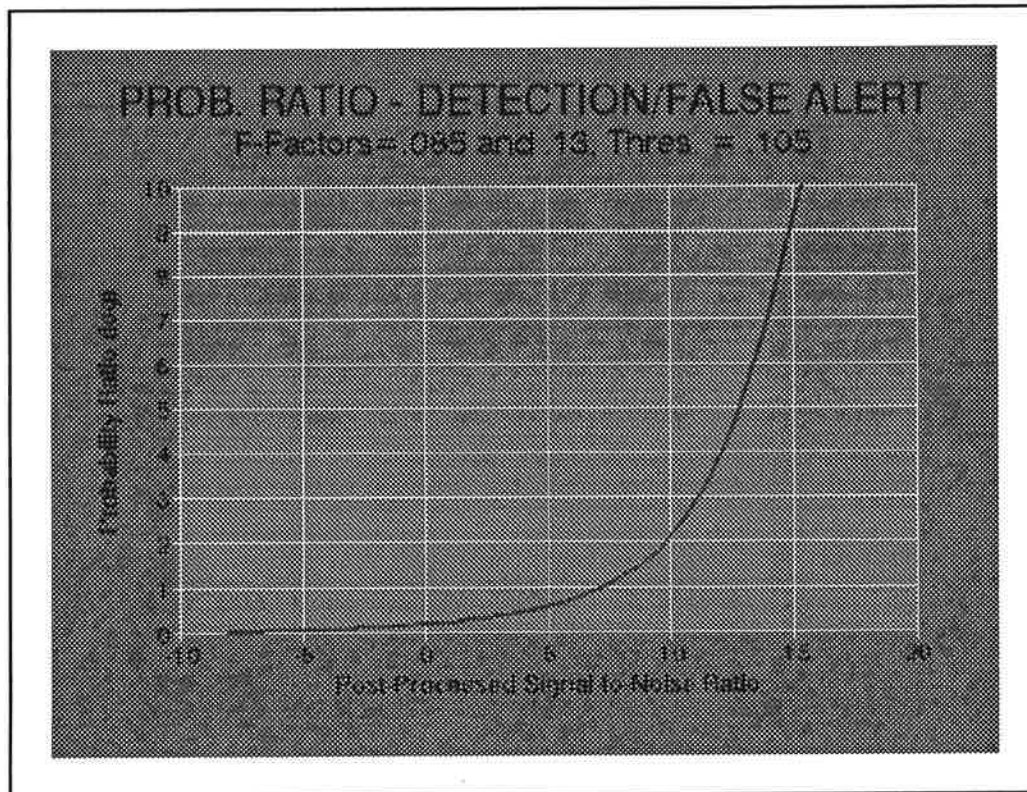


**FIGURE 8** DETECTION AND FALSE ALERT PROBABILITY PLOTTED VERSUS THE SINGLE PULSE SIGNAL-TO-NOISE RATIO.

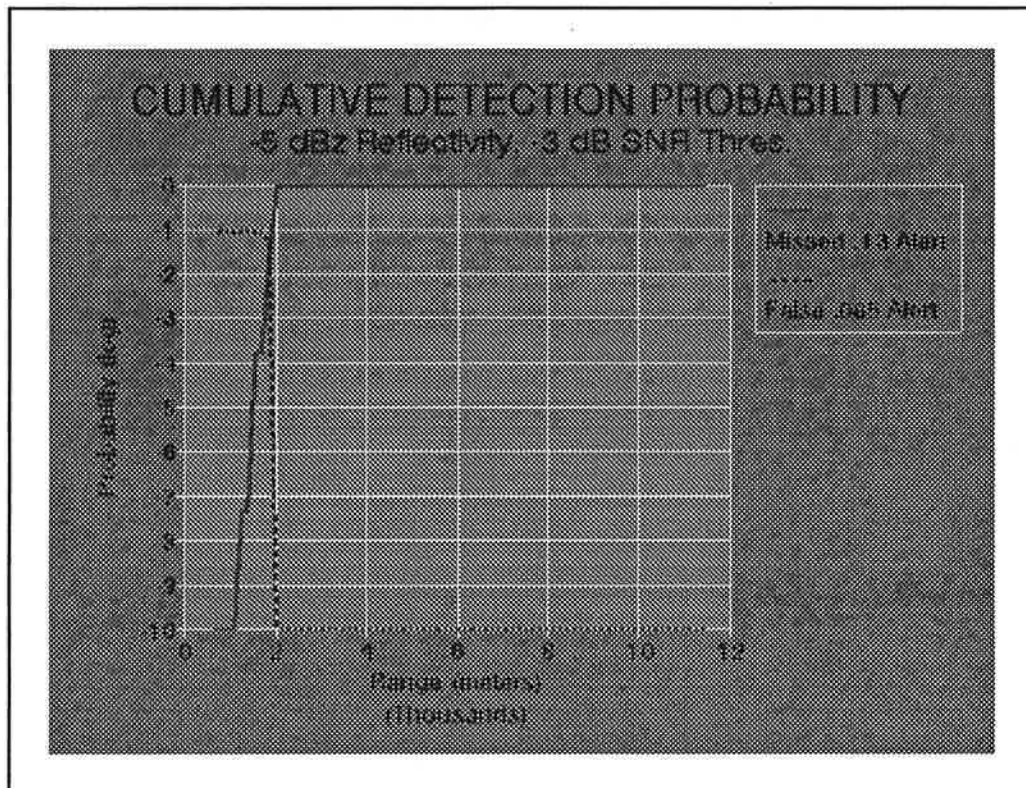




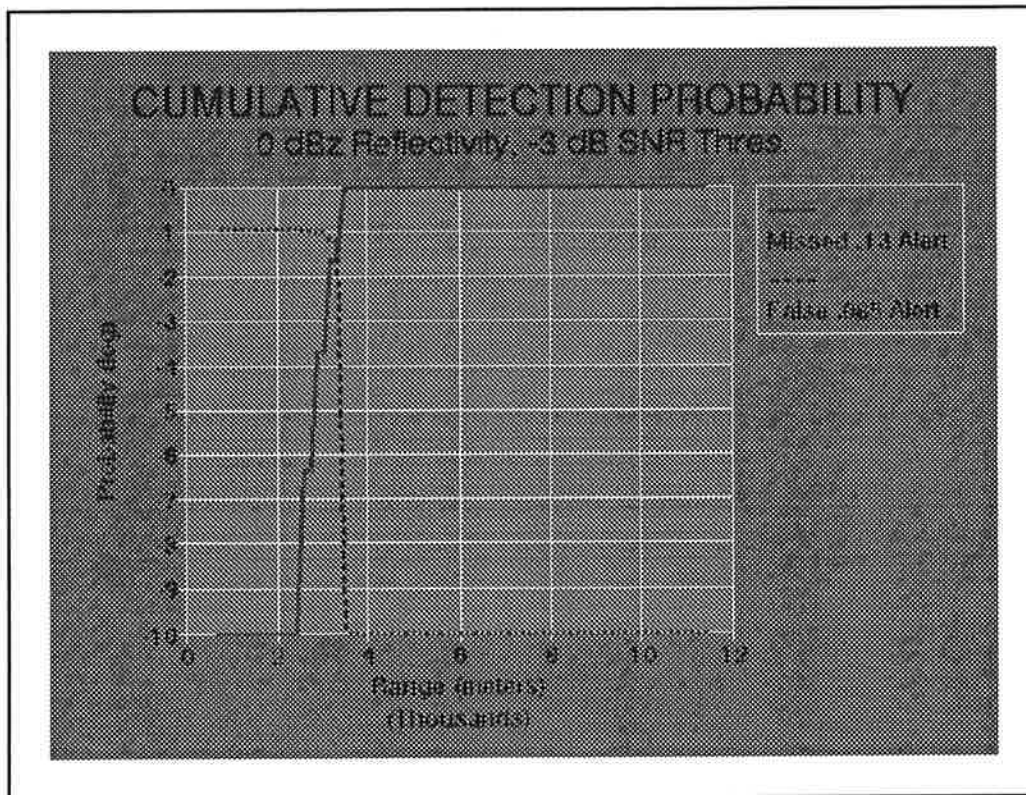
**FIGURE 9** RATIO OF DETECTION TO FALSE ALERT PROBABILITY PLOTTED VERSUS SINGLE PULSE SIGNAL-TO-NOISE RATIO.



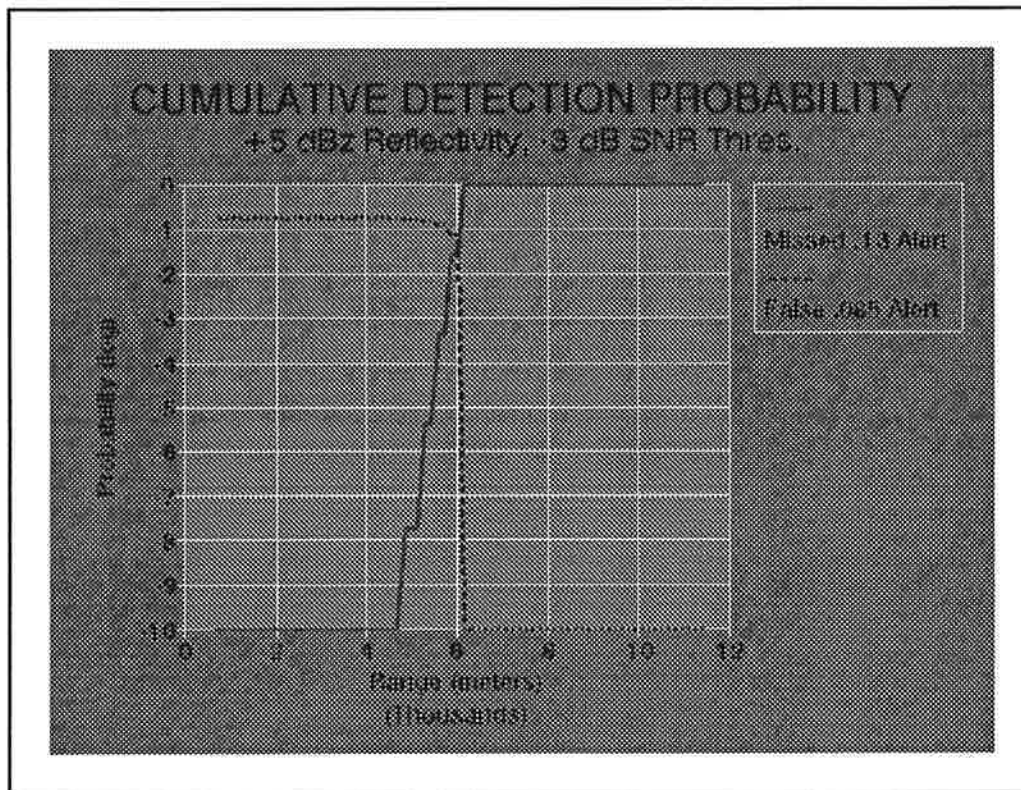
**FIGURE 10** RATIO PLOT SIMILAR TO FIGURE 9 EXCEPT THAT THE POSTPROCESSED SIGNAL-TO-NOISE RATIO IS USED AS THE ORDINATE. A SIGNAL-TO-NOISE RATIO THRESHOLD OF APPROXIMATELY 7.5 DB MUST BE USED TO MAINTAIN THE RATIO OF ONE FALSE ALERT PER 10 TRUE ALERTS.



**FIGURE 11** CUMULATIVE PROBABILITY OF MISSED AND FALSE DETECTION FOR -5 DBZ WEATHER REFLECTIVITY AND AN SNR THRESHOLD OF -3DB.

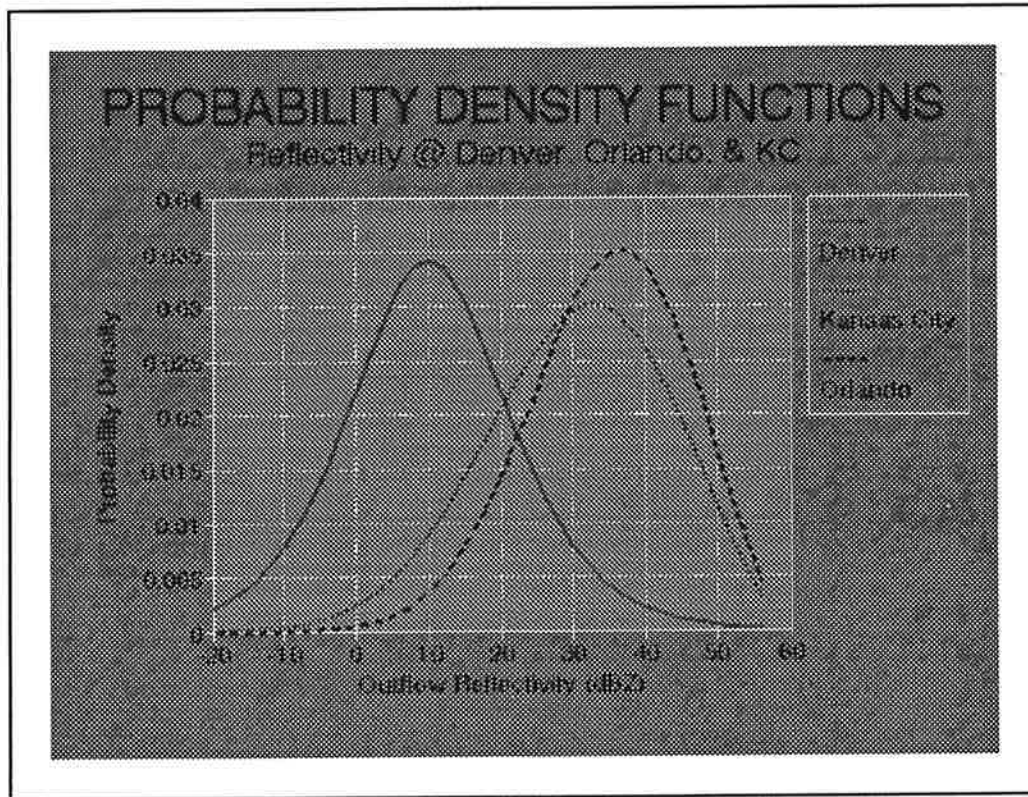


**FIGURE 12** CUMULATIVE PROBABILITY OF MISSED AND FALSE DETECTION FOR -0 DBZ WEATHER REFLECTIVITY AND AN SNR THRESHOLD OF -3 DB.

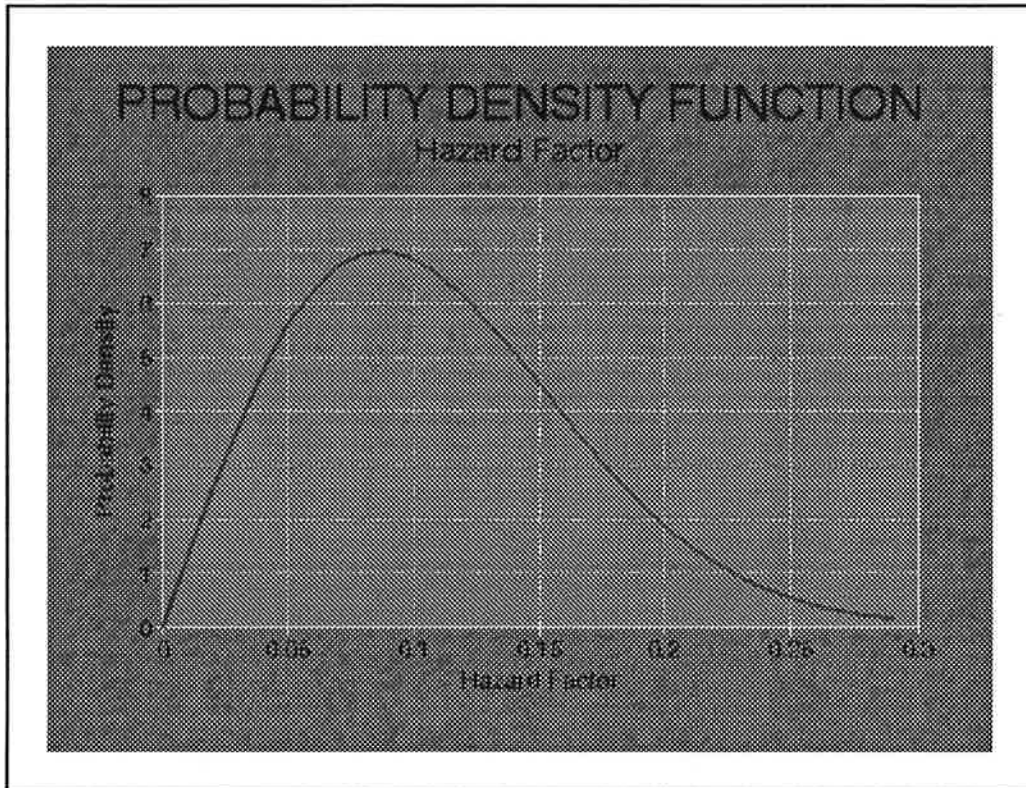


**FIGURE 13** CUMULATIVE PROBABILITY OF MISSED AND FALSE DETECTION FOR +5 DBZ WEATHER REFLECTIVITY AND AN SNR THRESHOLD OF -3 DB.

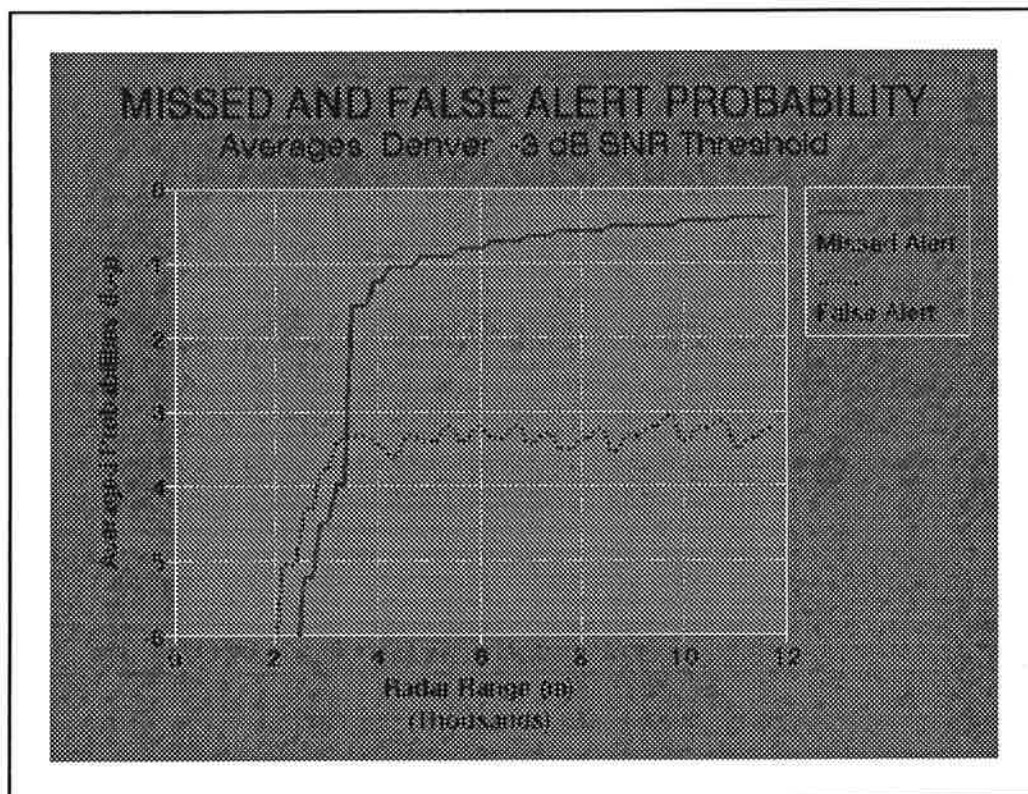




**FIGURE 14** PLOT OF THE PROBABILITY DENSITY FUNCTIONS OF OUTFLOW REFLECTIVITY FROM DENVER, ORLANDO AND KANSAS CITY (FROM REFERENCE 1 & 5).

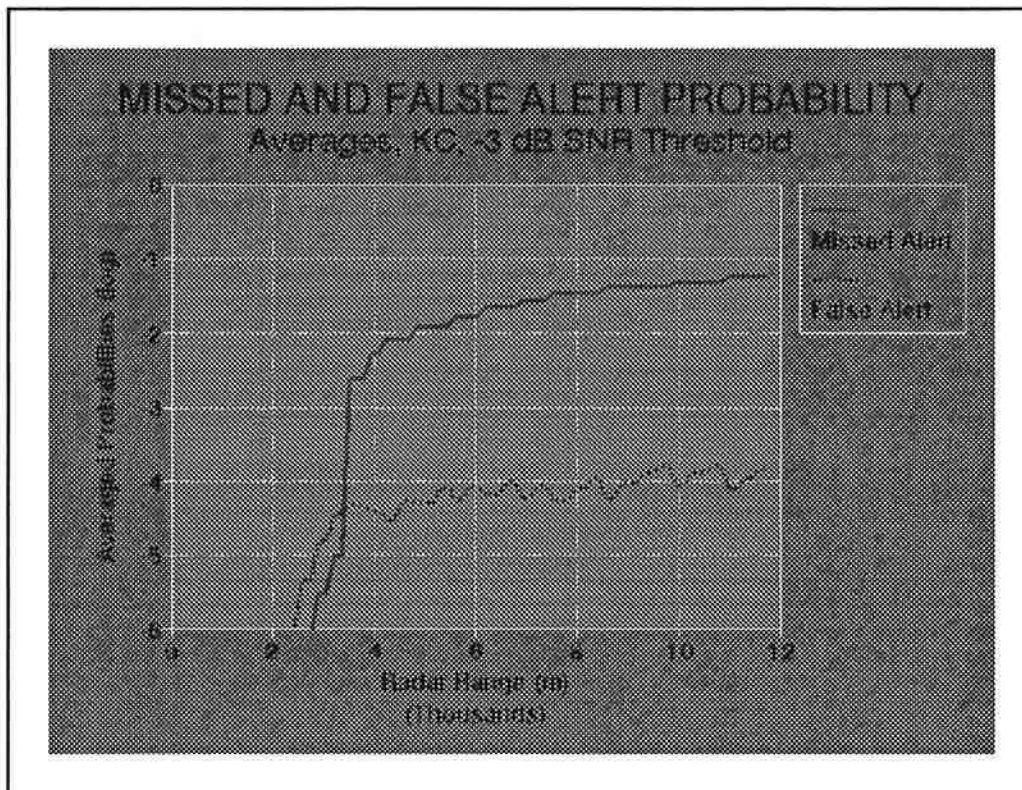


**FIGURE 15** PLOT OF THE PROBABILITY DENSITY FUNCTION OF HAZARD FACTOR (FROM REFERENCE 1). IT IS ASSUMED THAT THIS DENSITY FUNCTION APPLIES TO THE 1000 METER AVERAGED F-FACTOR.

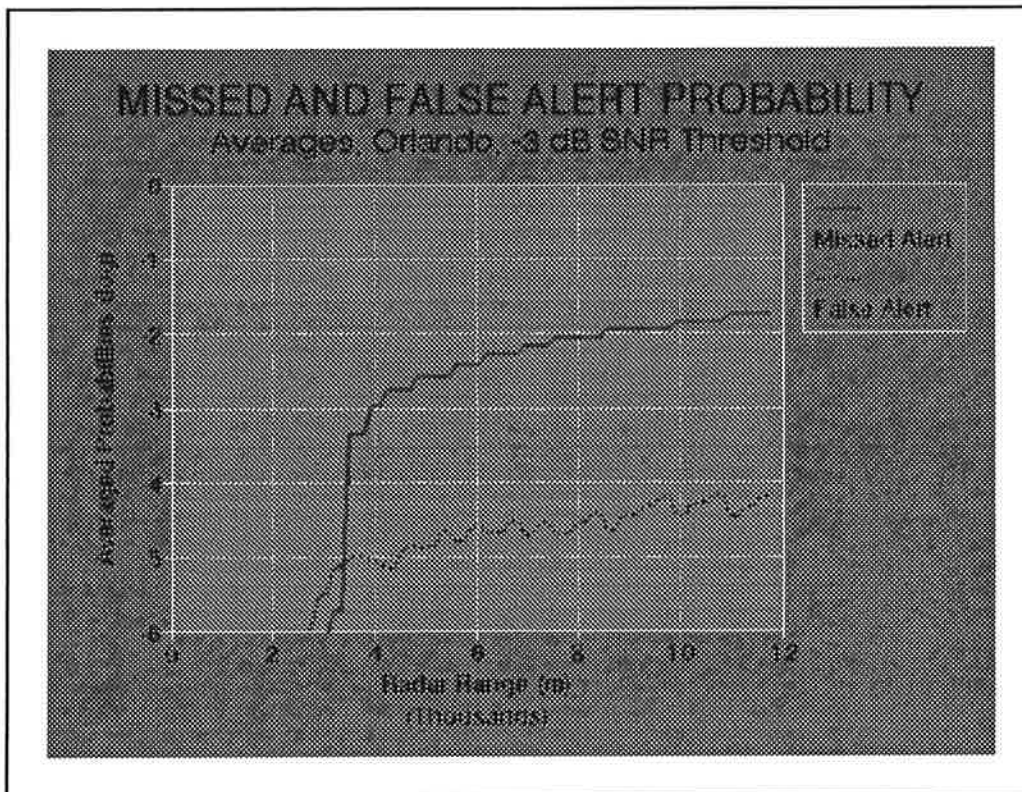


**FIGURE 16** AVERAGED PROBABILITIES OF A MISSED AND FALSE ALERT ON A SINGLE SCAN AT DENVER FOR MICRO-BURSTS EXCEEDING 0 DBZ REFLECTIVITY.

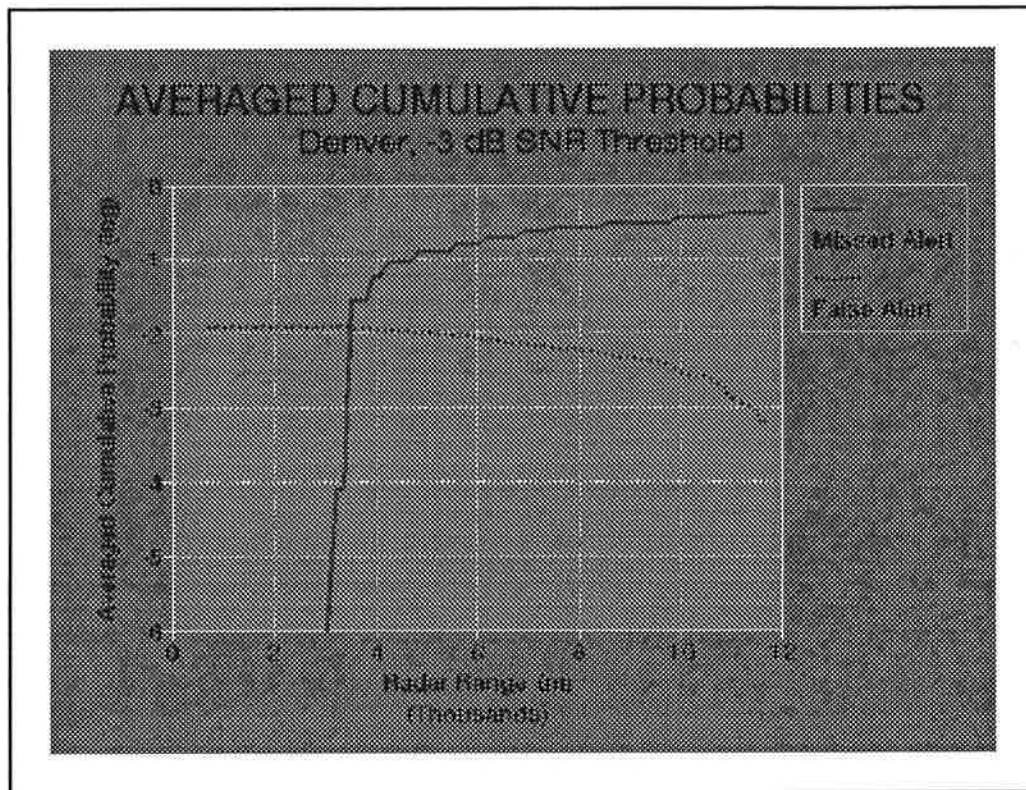




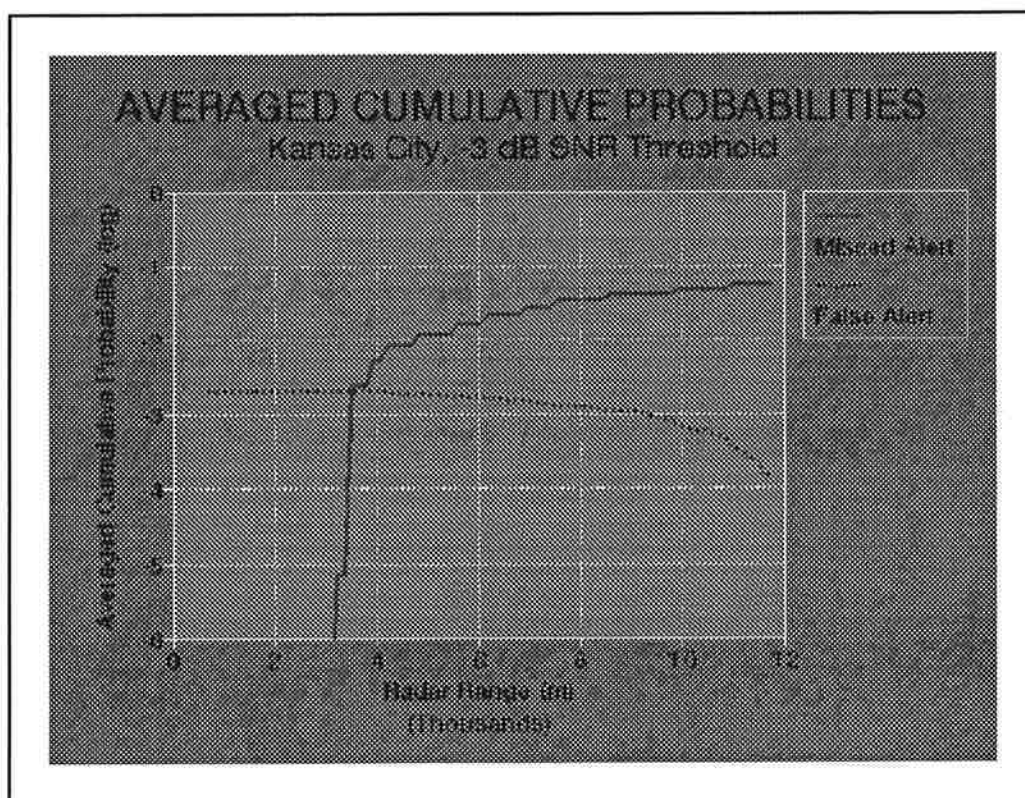
**FIGURE 17 - AVERAGED PROBABILITIES OF A MISSED AND FALSE ALERT ON A SINGLE SCAN AT KANSAS CITY FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY.**



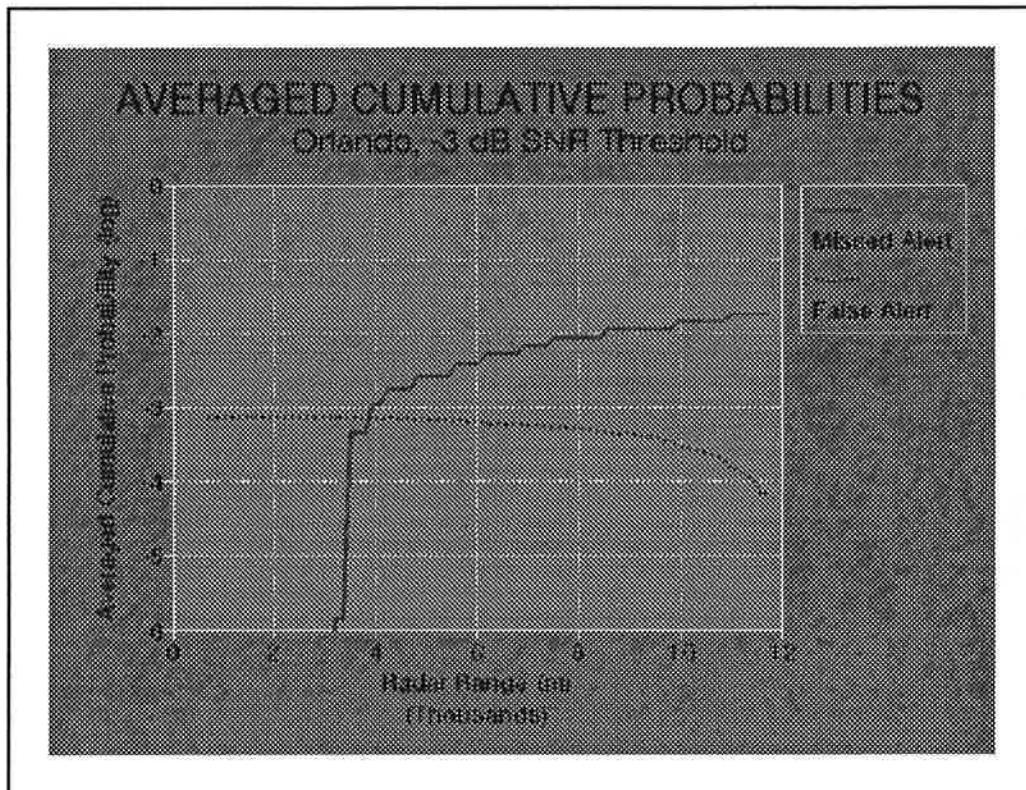
**FIGURE 18** AVERAGED PROBABILITIES OF A MISSED AND FALSE ALERT ON A SINGLE SCAN AT ORLANDO FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY.



**FIGURE 19** AVERAGED CUMULATIVE PROBABILITIES OF A MISSED AND FALSE ALERT AT DENVER USING AN SNR THRESHOLD (SINGLE-PULSE) OF -3 DB FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY.

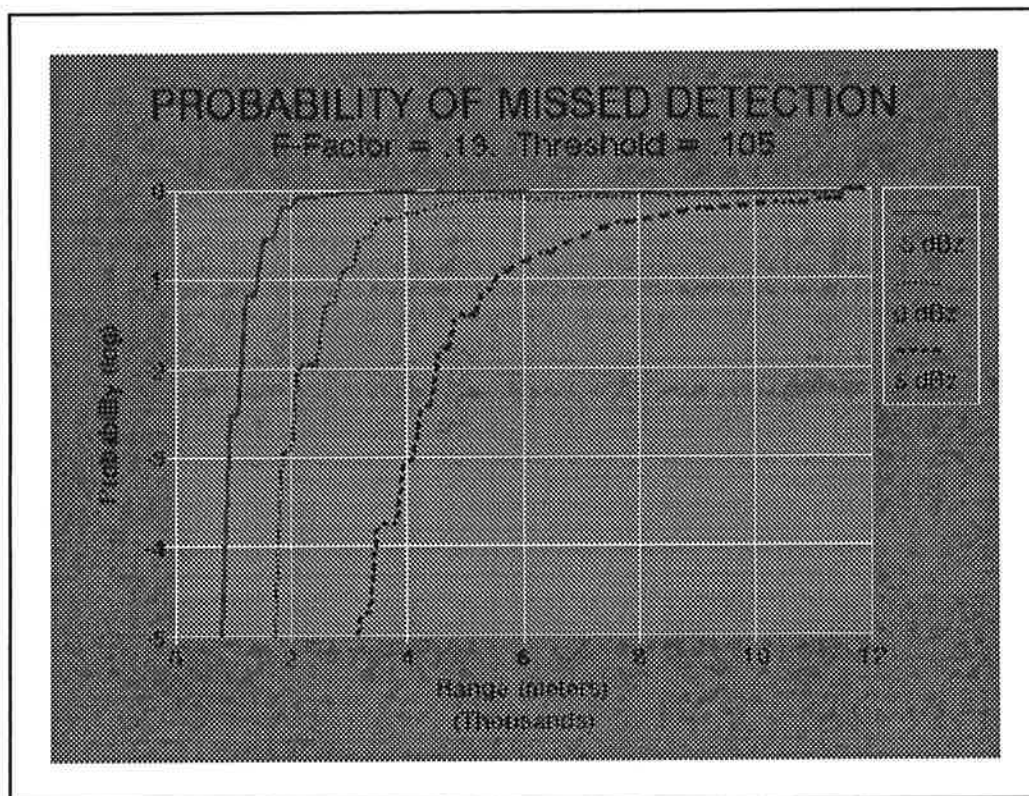


**FIGURE 20** AVERAGED CUMULATIVE PROBABILITIES OF A MISSED AND FALSE ALERT AT KANSAS CITY USING AN SNR THRESHOLD (SINGLE-PULSE) OF -3 DB FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY.

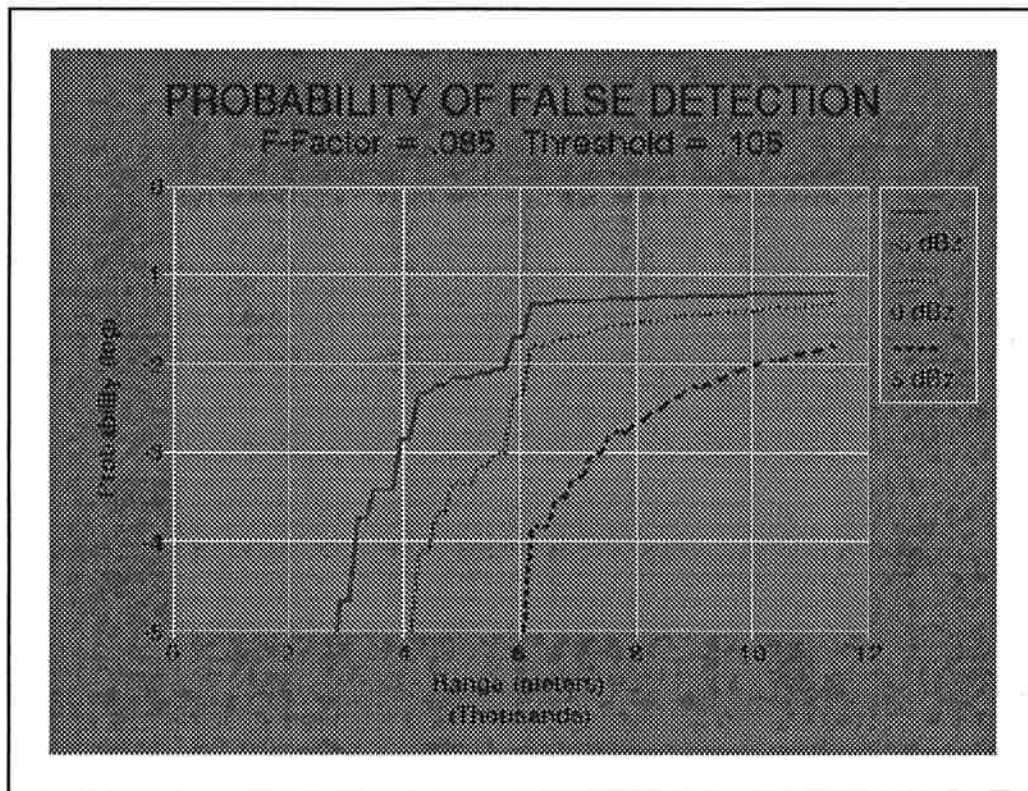


**FIGURE 21** AVERAGED CUMULATIVE PROBABILITIES OF A MISSED AND FALSE ALERT AT ORLANDO USING AN SNR THRESHOLD (SINGLE-PULSE) OF -3 DB FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY.

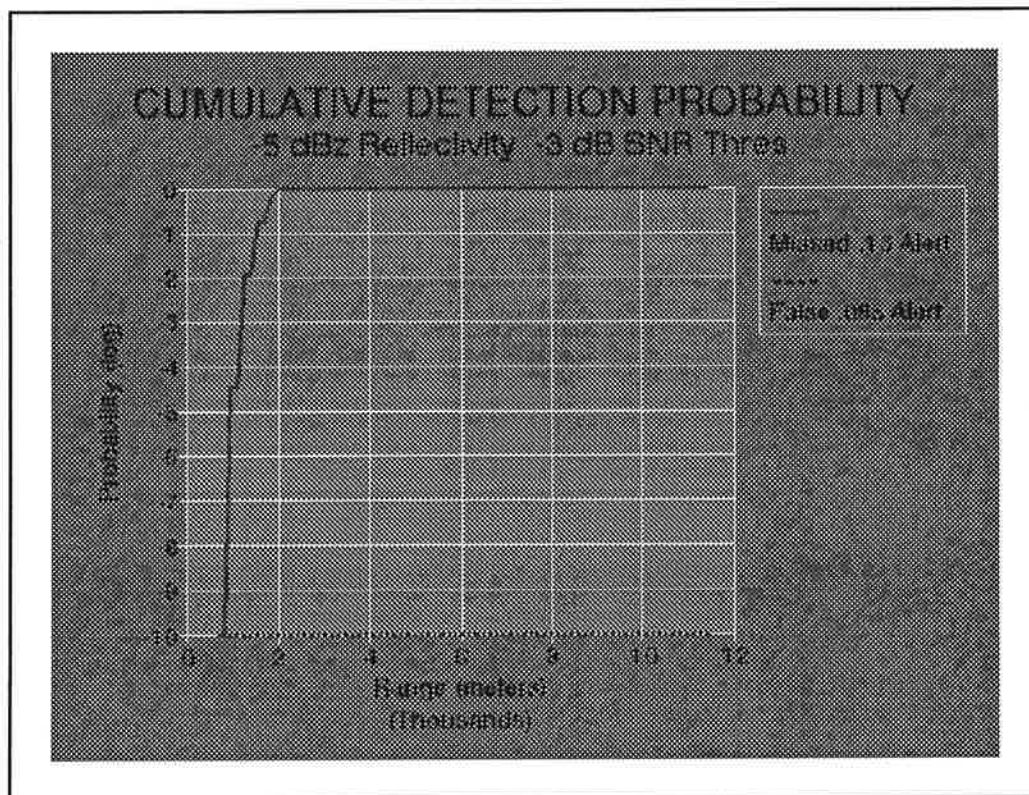




**FIGURE 22** PROBABILITY OF A MISSED DETECTION OF A .13 HAZARD FOR THE NASA SYSTEM USING A HAZARD AREA THRESHOLD OF .2 SQ. KM. AND REQUIRING TWO CONSECUTIVE SCANS PRIOR TO DECLARING AN ALERT.

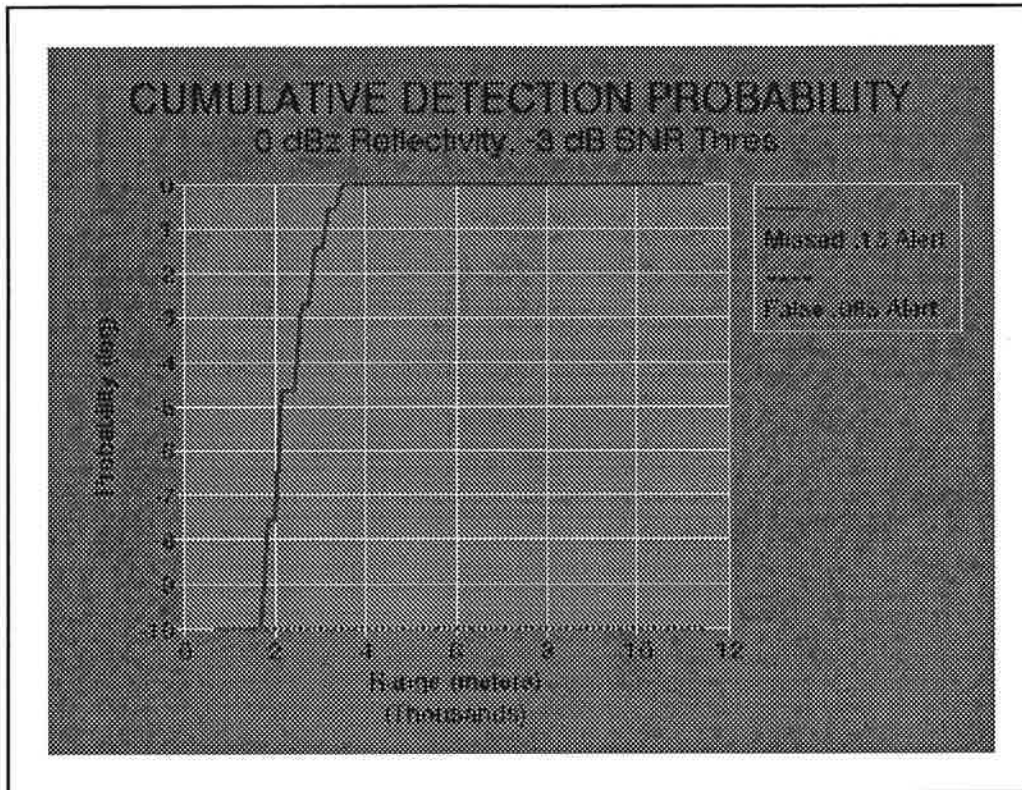


**FIGURE 23** PROBABILITY OF A FALSE DETECTION OF A .085 HAZARD FOR THE NASA SYSTEM USING A HAZARD AREA THRESHOLD OF .2 SQ. KM. AND REQUIRING TWO CONSECUTIVE SCANS PRIOR TO DECLARING AN ALERT.

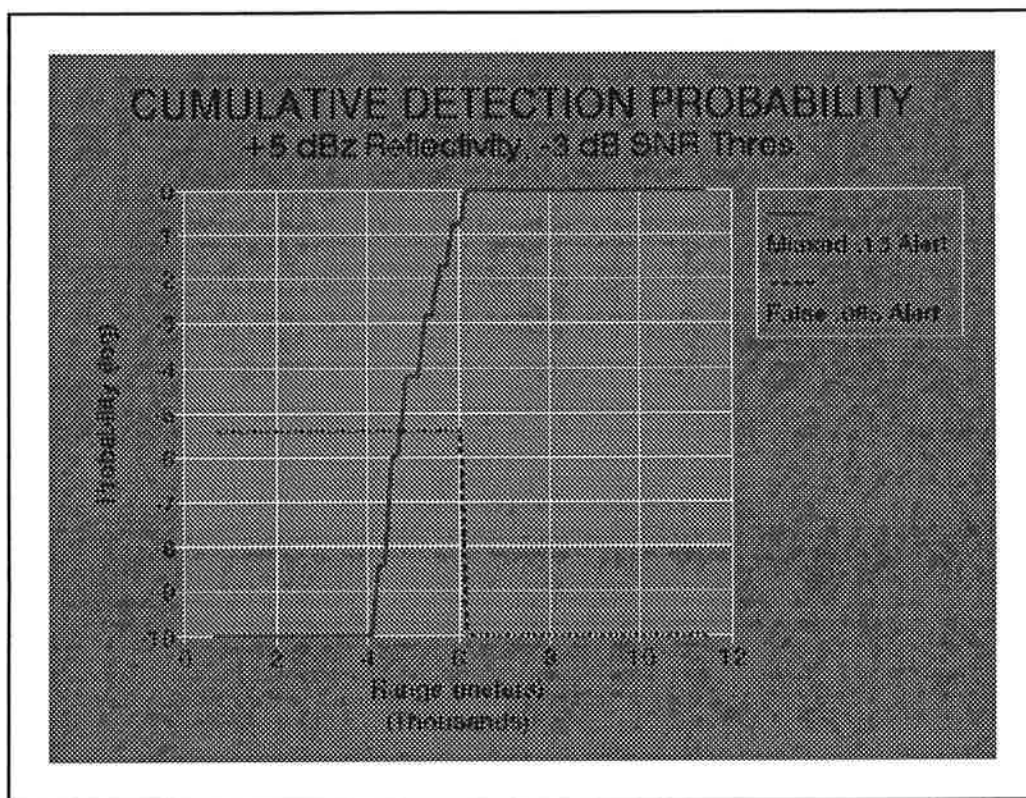


**FIGURE 24** CUMULATIVE PROBABILITY OF A MISSED AND FALSE DETECTION FOR THE NASA SYSTEM WITH AN SNR THRESHOLD OF -3 DB AND A WEATHER REFLECTIVITY OF -5 DBZ.

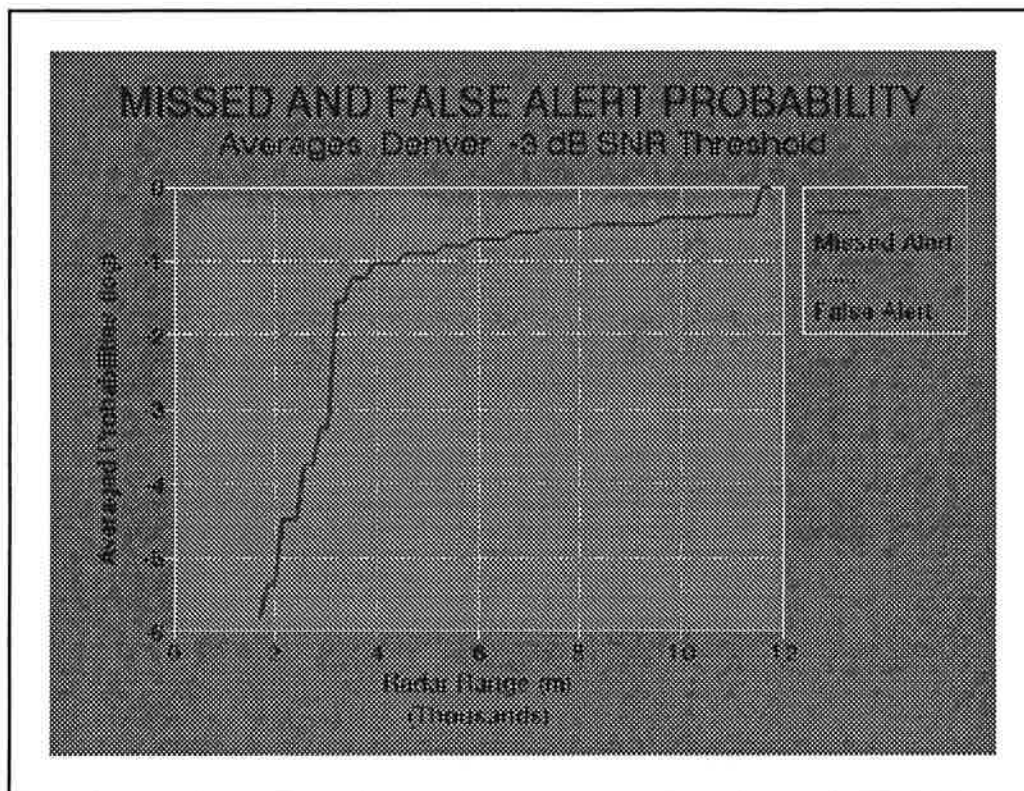




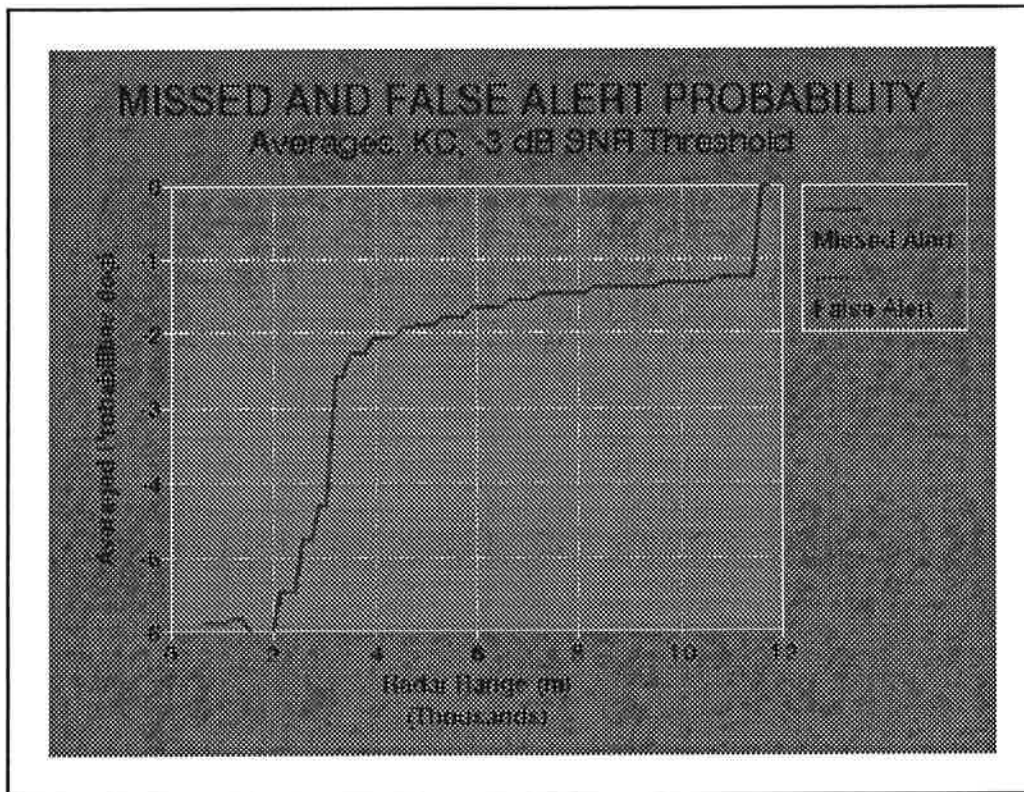
**FIGURE 25** CUMULATIVE PROBABILITY OF A MISSED AND FALSE DETECTION FOR THE NASA SYSTEM WITH AN SNR THRESHOLD OF -3 DB AND A WEATHER REFLECTIVITY OF -0 DBZ.



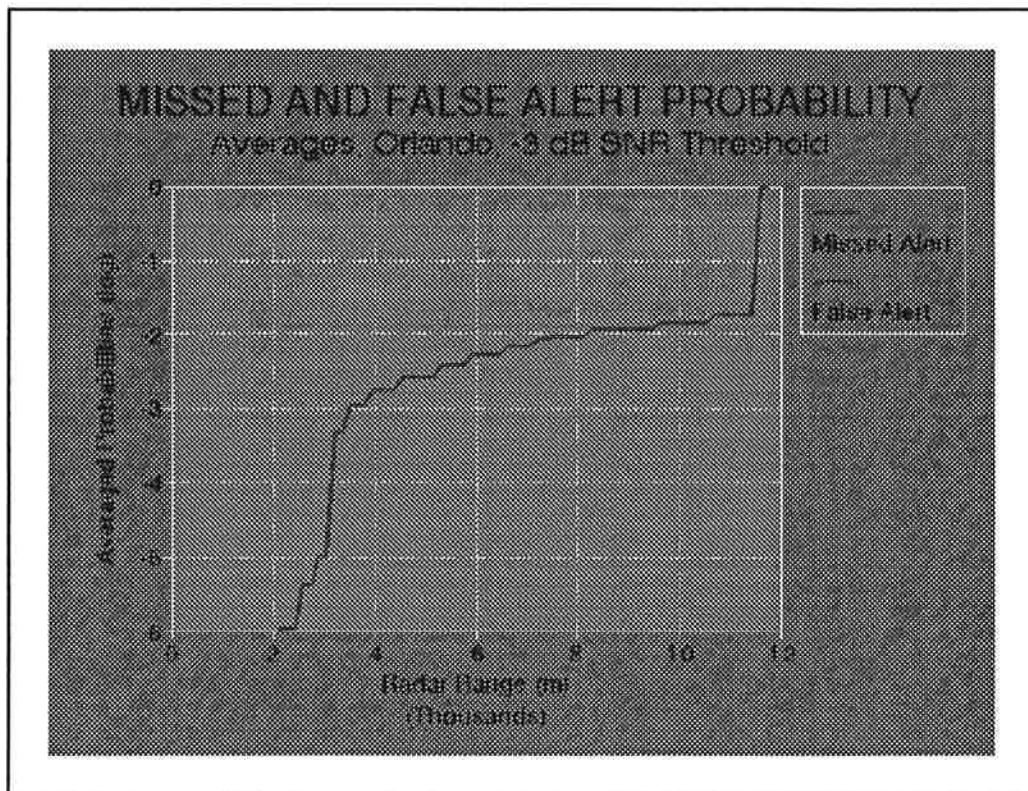
**FIGURE 26** CUMULATIVE PROBABILITY OF A MISSED AND FALSE DETECTION FOR THE NASA SYSTEM WITH AN SNR THRESHOLD OF -3 DB AND A WEATHER REFLECTIVITY OF +5 DBZ.



**FIGURE 27** AVERAGED PROBABILITY OF MISSED AND FALSE DETECTION AT DENVER FOR THE NASA SYSTEM WITH HAZARD AREA THRESHOLD AND MULTIPLE SCAN DETECTION FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).

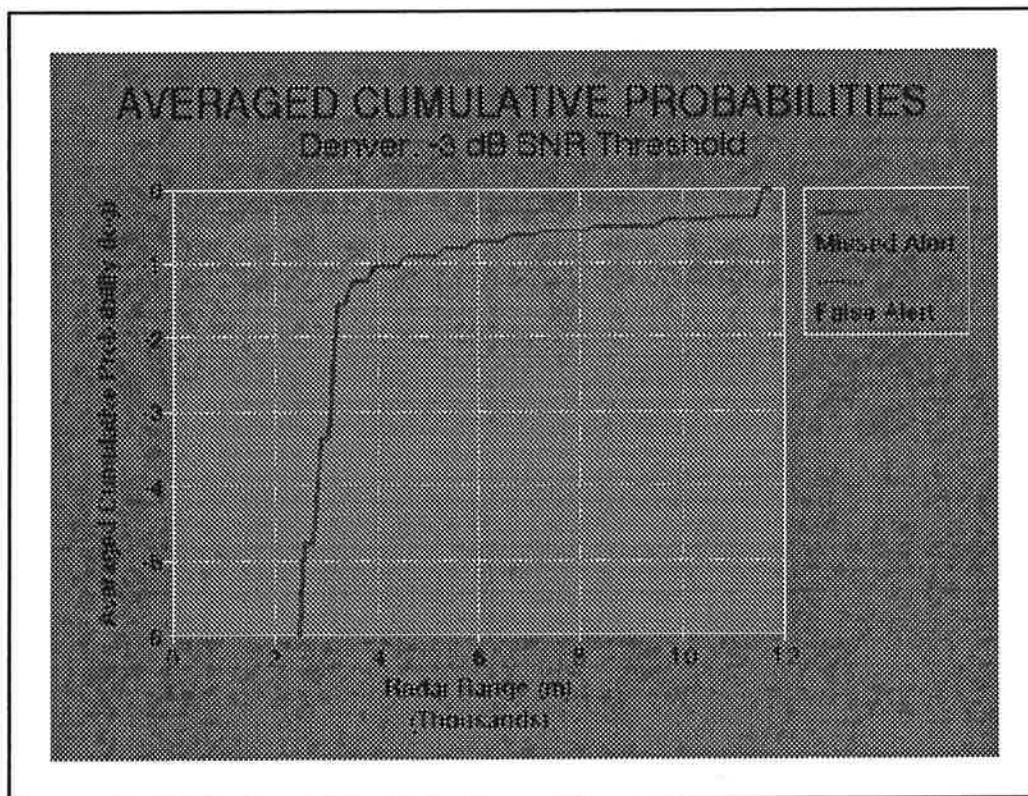


**FIGURE 28** AVERAGED PROBABILITY OF MISSED AND FALSE DETECTION AT KANSAS CITY FOR THE NASA SYSTEM WITH HAZARD AREA THRESHOLD AND MULTIPLE SCAN DETECTION FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).

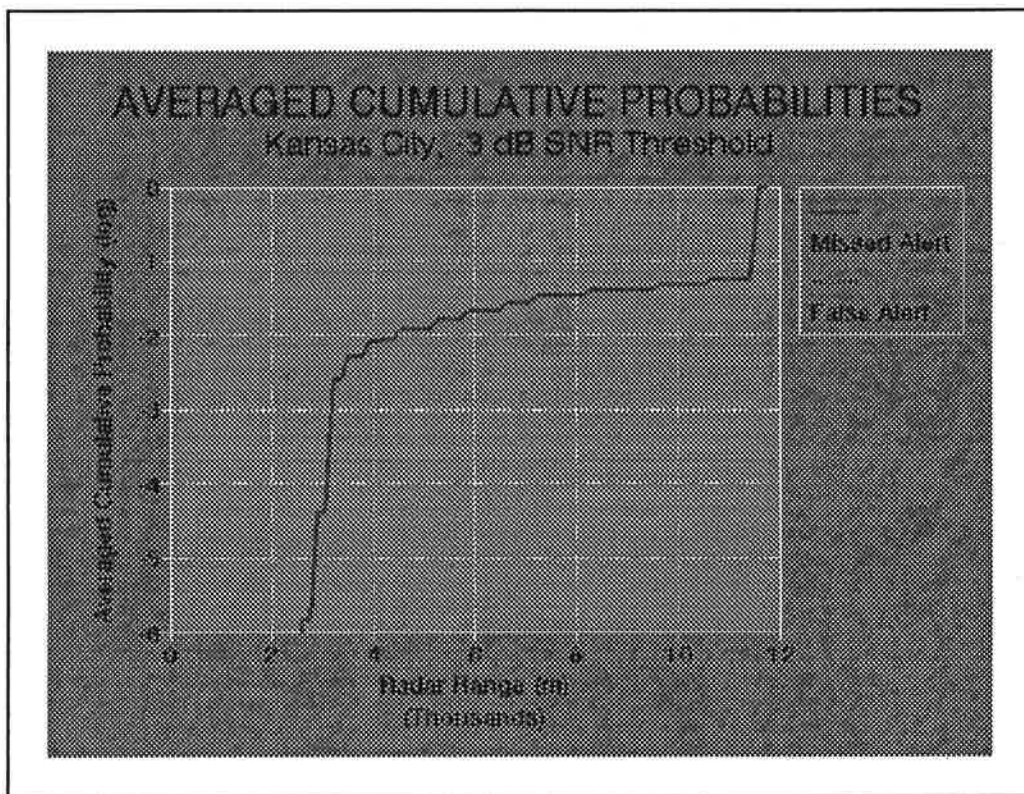


**FIGURE 29** AVERAGED PROBABILITY OF MISSED AND FALSE DETECTION AT ORLANDO FOR THE NASA SYSTEM WITH HAZARD AREA THRESHOLD AND MULTIPLE SCAN DETECTION FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).

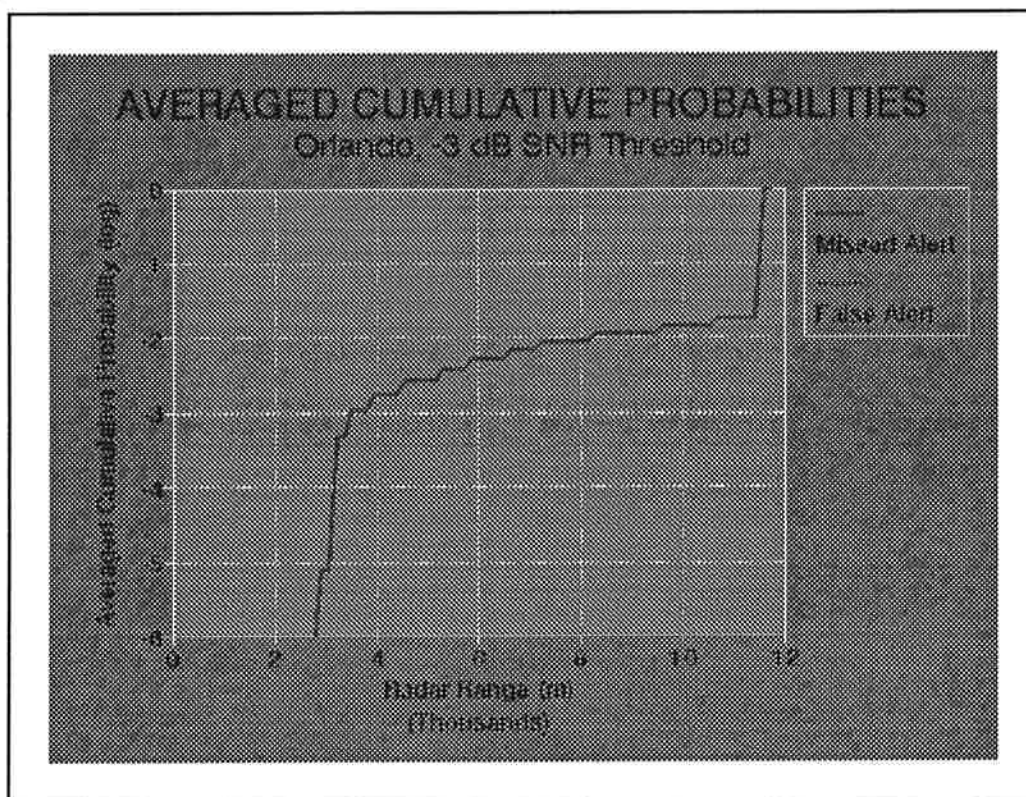




**FIGURE 30** AVERAGED CUMULATIVE PROBABILITY OF MISSED AND FALSE DETECTION AT DENVER FOR THE NASA SYSTEM WITH HAZARD AREA THRESHOLD AND MULTIPLE SCAN DETECTION FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).



**FIGURE 31** AVERAGED CUMULATIVE PROBABILITY OF MISSED AND FALSE DETECTION AT KANSAS CITY FOR THE NASA SYSTEM WITH HAZARD AREA THRESHOLD AND MULTIPLE SCAN DETECTION FOR MICROBURSTS EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).



**FIGURE 32** AVERAGED CUMULATIVE PROBABILITY OF MISSED  
2ND AND FALSE DETECTION AT ORLANDO FOR THE NASA  
SYSTEM WITH HAZARD AREA THRESHOLD AND  
MULTIPLE SCAN DETECTION FOR MICROBURSTS  
EXCEEDING 0 DBZ REFLECTIVITY. THE PLOT OF FALSE  
ALERT PROBABILITY IS LESS THAN  $10^{-6}$  (OFF SCALE).



## REFERENCES

- [1] Mathews, B. D. and Fran Miller: "Principles and Computation of Microburst Windshear Hazard Factor Detection Range Performance with Airborne Pulsed Doppler Radar". Westinghouse Electric Corporation, Baltimore, Maryland (Westinghouse Proprietary preliminary paper).
- [2] Doviak, R. J. and D. S Zrnic: Doppler Radar and Weather Observations, Academic Press, Inc., New York.
- [3] Wonnacott, T. H. and R. J. Wonnacott. Introductory Statistics, John Wiley & Sons, New York.
- [4] Bowles, R. L. *Windshear Detection and Avoidance: Airborne Systems Survey*, Proceedings of the 29th Conference on Decision and Control. Honolulu, Hawaii, December 1990.
- [5] Campbell, S. M.I.T., Lincoln Laboratory, personal communication